

3

Aerial Image Formation – The Details

The impact of aberrations and defocus must now be added to the description of image formation provided in the previous chapter. The unique aspects of imaging while scanning the mask and wafer past the stationary imaging lens will also be included. Next, a discussion of the vector nature of light and the impact of polarization on imaging will be added and immersion lithography will be described. Finally, a preliminary discussion of image quality will conclude this chapter.

3.1 Aberrations

According to Webster, an aberration is ‘. . . a departure from what is right, true or correct’. In optical imaging, ‘right, true or correct’ can be thought of as the ideal, ‘diffraction-limited’ imaging performance of a lens (which was rigorously defined in the previous chapter using Fourier optics). Thus, a lens aberration is any deviation of the real performance of that lens from its ideal performance. As one might imagine, aberrations are undesirable intrusions of reality into our attempts to achieve imaging perfection.

3.1.1 The Causes of Aberrations

In practice, aberrations come from three sources – aberrations of design, aberrations of construction and aberrations of use. Aberrations of construction are probably the most tangible sources of errors and include incorrect shapes and thickness of the glass elements that make up the lens, inhomogeneous glass used in their construction, improper mounting, spacings or tilts of the various lens elements, or other imperfections in the manufacture of the lens (Figure 3.1). Aberrations of use include all ways in which improper use of the lens degrades its performance: using the wrong wavelength or wavelength

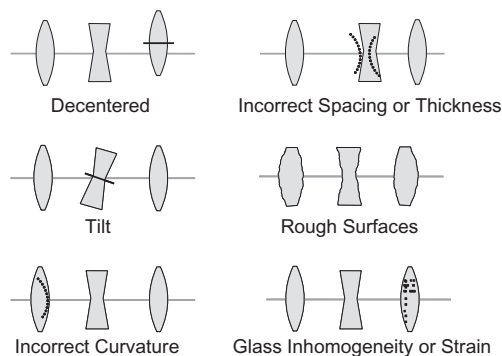


Figure 3.1 Examples of typical aberrations of construction

spectrum, tilt of the mask or wafer plane, or incorrect environmental conditions (e.g. changes in the refractive index of air due to changes in temperature, humidity or barometric pressure). But in a sense, it is the aberrations of design which are the fundamental problem for lithographic lenses.

An ‘aberration of design’ does not mean mistakes or problems caused by the designer of the lens. In fact, it means quite the opposite. Aberrations are fundamental to the nature of imaging and it is the goal of the lens designer to ‘design out’ as many of these aberrations as possible. The aberrations of design are those aberrations, inserted into the imaging system by nature, that the designer was not able to extract. The root cause of these aberrations for refractive optical elements is the nonlinear nature of Snell’s law.

Snell’s law describes the refraction of light as it passes from one material to another. Light incident at the boundary of two materials at an angle θ_1 with respect to the normal will refract to a new angle θ_2 in material 2 given by Snell’s law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (3.1)$$

where n_1 and n_2 are the refractive indices of the two materials. It is this refraction between air and glass that allows a curved glass surface to focus light. In terms of refraction, the goal of imaging can be stated very simply: light emanating in all directions from some point on the object should be collected by the lens and focused to its ideal image point. Thus, a ray of light coming from the object point should pass through the lens and arrive at the image point, regardless of its angle.

Does Snell’s law permit this ideal behavior? For a lens made of a single element (i.e. a single piece of glass) with spherical front and back surfaces, it is not difficult to show that ideal imaging behavior would be obtained *only* if Snell’s law were linear: $n_1 \theta_1 = n_2 \theta_2$. Thus, it is the nonlinear nature of the refraction of light that causes nonideal imaging behavior. In the limit of small angles, Snell’s law does behave about linearly ($\sin \theta \approx \theta$ for small θ). When all the angles of light passing through the imaging system are small enough so that Snell’s law is approximately linear, we find that the lens behaves ideally in this *paraxial* (near the axis) approximation. Although a spherically shaped lens was used as an example, there is no single shape for a lens surface that provides ideal imaging for many different object points. Thus, fundamentally, the nonlinear nature of Snell’s law results in nonideal imaging behavior (i.e. aberrations) for any single lens element.

If aberrations are fundamental to the nature of imaging, how can we make a lens that approaches the ideal imaging behavior? Since each lens element has aberrations by its very nature, these aberrations are corrected by combining different lens elements with differing aberrations. If two lens elements are brought together that have aberrations of similar magnitude but in opposite directions, the combination will have smaller aberrations than either individual lens element. A simple example is a pair of eyeglasses. One lens, your eye, has aberrations of a certain magnitude and direction. A second lens, in the glasses, is designed to have aberrations of about the same magnitude but in the opposite direction. The combination yields residual aberrations that are small enough to be acceptable. One of the simplest examples of aberration correction through lens design is a cemented doublet of positive and negative elements made from two different types of glass, which has the added advantage of reducing chromatic aberrations – the variation of imaging behavior with wavelength (Figure 3.2).

The combination of two lens elements can reduce the aberrations of the overall lens system, but not eliminate them. If a third element is added, the total amount of aberrations can be reduced further. Each lens element, with its two curved surfaces, thickness, position and type of glass, adds more degrees of freedom to the design to allow the overall level of aberrations to be reduced. Thus, the lens designer uses these degrees of freedom (in addition to the position of an aperture stop) to ‘design out’ the aberrations that nature has embedded into the behavior of a single element (see Figure 3.3 for an example of a modern lithographic lens). An aberration free lens system, however, is an ideal that can never be achieved in practice. It would take an infinite number of lens elements to make the designed behavior perfect. Before this point is reached, however, aberrations of construction would begin to dominate. The optimum number of elements is reached when the addition of one more lens element adds more aberrations of construction than it reduces the aberrations of design.

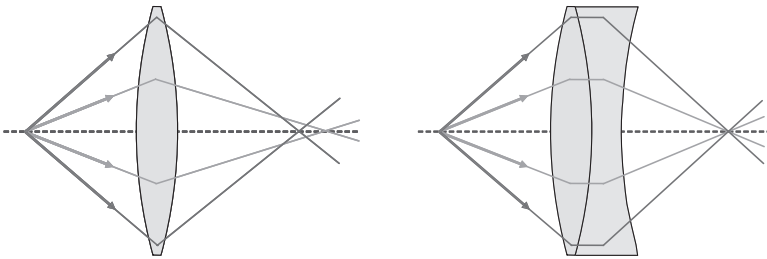


Figure 3.2 The reduction of spherical aberration by the use of a cemented doublet

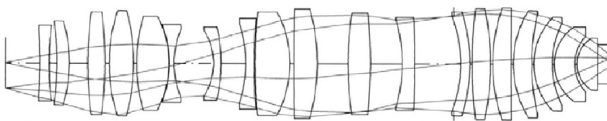


Figure 3.3 Example of a simple $NA = 0.8$, 248-nm lens design¹

Since the nonlinear nature of Snell's law leads to aberrations, larger angles passing through the lens produce greater nonideal behavior. Thus, for a given lens design, increases in numerical aperture (NA) or field size result in large increases in aberrations. In fact, however, new generations of lithographic lenses with higher NAs always seem to exhibit *lower* levels of aberrations compared to previous generations due to improvements in lens design and manufacturing. The increasing use of aspherical lens elements, with their greater degrees of freedom for the lens design, has helped to keep the number of lens elements lower than would be otherwise required (though at a cost of increased manufacturing complexity). Still, lens size and weight grow considerably for each generation of lithographic tool, with 500-kg lenses more than 2 m tall now common.

Unfortunately, even the low levels of aberrations found in a state-of-the-art lens are not small enough to be ignored. Invariably, the high NA lens is used to print ever smaller features that are increasingly sensitive to even the slightest deviation from perfection. The impact of these aberrations is a reduction of image quality, and ultimately a loss of critical dimension (CD) control and overlay performance (see Chapter 8). A thorough understanding of the aberration behavior of a lens is a necessity for those seeking to push the limits of their lithography tools.

3.1.2 Describing Aberrations: the Zernike Polynomial

How are aberrations characterized? In the simple geometrical optics approach, optical *rays* represent the propagation of light through the optical system. By tracing the paths of the light through a lens (using a lens design software program, for example), the deviations in the actual paths of the rays from the ideal paths tell us the magnitude of the aberration (Figure 3.4). The magnitude of the aberration can be expressed as an optical path difference (OPD) between the ideal and actual rays. Rays point in the direction that light is traveling and are thus perpendicular to the *wavefront* of the light, a surface of constant phase. By propagating these wavefronts through the lens, the deviation of the ideal wavefront from the real wavefront can also be used to express the magnitude of the

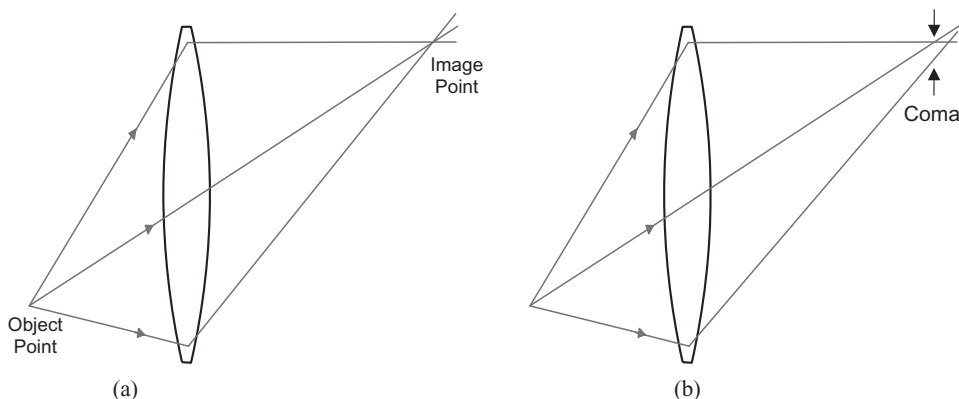


Figure 3.4 Ray tracing shows that (a) for an ideal lens, light coming from the object point will converge to the ideal image point for all angles, while (b) for a real lens, the rays do not converge to the ideal image point

aberration (Figure 3.5). The two descriptions are completely equivalent since light undergoes a phase change of 2π every time it travels a distance of one wavelength, providing a simple conversion between OPD and phase error.

The aberration behavior of a lens can be predicted using lens design software, or measured using interferometry. In both cases, the result is a map of the phase error (or OPD) of the light exiting from the lens for a given point in the field. It is important to note that the aberrations will vary as a function of field position (center of the field versus the upper left corner, for example). It typically takes 25 to 50 measurements across the field to fully characterize how these aberrations vary for a typical lithographic lens.

A map of the phase error across the exit pupil of the lens is a useful, but potentially large, set of data. One of the most common ways to better understand such data is to curve fit the data to a function, typically a high-order polynomial. In that way, this large set of experimental data can be represented by relatively few polynomial coefficients. Further, by carefully choosing the form of the polynomial fitting function, the coefficients themselves can take on physically meaningful interpretations. By far the most common such polynomial is the *Zernike polynomial*, named for Frits Zernike, the 1953 Nobel prize winning physicist. This orthogonal polynomial series has an infinite number of terms, but is typically cut off after 36 terms, with powers of the relative radial pupil position R and trigonometric functions of the polar angle ϕ . Thus, knowing the numerical values of the 36 Zernike coefficients allows one to describe to a reasonable level of accuracy the aberration behavior of a lens at a single field point.

The Zernike polynomial can be arranged in many ways, but most lens design software and lens measuring equipment employ the fringe or circle Zernike polynomial, defined as

$$W(R, \phi) = \frac{OPD}{\lambda} = \sum_{i=0}^{\infty} Z_i F_i(R, \phi) \quad (3.2)$$

where $W(R, \phi)$ is the optical path difference relative to the wavelength, Z_i is called the i th Zernike coefficient, and $F_i(R, \phi)$ are the polynomial terms, defined in Table 3.1. The

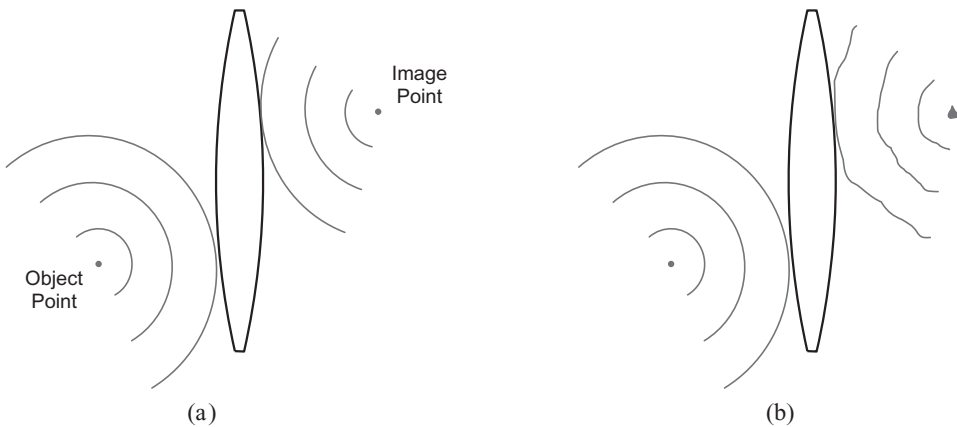


Figure 3.5 Wavefronts showing the propagation of light (a) for an ideal lens and (b) for a lens with aberrations

phase error due to aberrations will be simply $2\pi W(R, \phi)$. The polar coordinates on the unit circle (R, ϕ) are related to the Cartesian spatial frequency coordinates by

$$f_x = \frac{NA}{\lambda} R \cos \phi, \quad f_y = \frac{NA}{\lambda} R \sin \phi \quad (3.3)$$

Table 3.1 Polynomial functions and common names for the first 36 terms in the Fringe Zernike polynomial

Term	Fringe Zernike Formula, F_i	Common Name
Z_0	1	Piston
Z_1	$R \cos \phi$	x-Tilt
Z_2	$R \sin \phi$	y-Tilt
Z_3	$2R^2 - 1$	Power (paraxial focus)
Z_4	$R^2 \cos 2\phi$	3rd Order Astigmatism
Z_5	$R^2 \sin 2\phi$	3rd Order 45° Astigmatism
Z_6	$(3R^2 - 2)R \cos \phi$	3rd Order x-Coma
Z_7	$(3R^2 - 2)R \sin \phi$	3rd Order y-Coma
Z_8	$6R^4 - 6R^2 + 1$	3rd Order Spherical
Z_9	$R^3 \cos 3\phi$	Trefoil (3rd Order 3-Point)
Z_{10}	$R^3 \sin 3\phi$	45° Trefoil
Z_{11}	$(4R^2 - 3)R^2 \cos 2\phi$	5th Order Astigmatism
Z_{12}	$(4R^2 - 3)R^2 \sin 2\phi$	5th Order 45° Astigmatism
Z_{13}	$(10R^4 - 12R^2 + 3)R \cos \phi$	5th Order x-Coma
Z_{14}	$(10R^4 - 12R^2 + 3)R \sin \phi$	5th Order y-Coma
Z_{15}	$20R^6 - 30R^4 + 12R^2 - 1$	5th Order Spherical
Z_{16}	$R^4 \cos 4\phi$	Quadrafoil (3rd Order 4-Point)
Z_{17}	$R^4 \sin 4\phi$	45° Quadrafoil
Z_{18}	$(5R^2 - 4)R^3 \cos 3\phi$	5th Order Trefoil (5th Order 3-Point)
Z_{19}	$(5R^2 - 4)R^3 \sin 3\phi$	5th Order 45° Trefoil
Z_{20}	$(15R^4 - 20R^2 + 6)R^2 \cos 2\phi$	7th Order Astigmatism
Z_{21}	$(15R^4 - 20R^2 + 6)R^2 \sin 2\phi$	7th Order 45° Astigmatism
Z_{22}	$(35R^6 - 60R^4 + 30R^2 - 4)R \cos \phi$	7th Order x-Coma
Z_{23}	$(35R^6 - 60R^4 + 30R^2 - 4)R \sin \phi$	7th Order y-Coma
Z_{24}	$70R^8 - 140R^6 + 90R^4 - 20R^2 + 1$	7th Order Spherical
Z_{25}	$R^5 \cos 5\phi$	Pentafoil (3rd Order 5-Point)
Z_{26}	$R^5 \sin 5\phi$	45° Pentafoil
Z_{27}	$(6R^2 - 5)R^4 \cos 4\phi$	5th Order Quadrafoil (5th Order 4-Point)
Z_{28}	$(6R^2 - 5)R^4 \sin 4\phi$	5th Order 45° Quadrafoil
Z_{29}	$(21R^4 - 30R^2 + 10)R^3 \cos 3\phi$	7th Order Trefoil (7th Order 3-Point)
Z_{30}	$(21R^4 - 30R^2 + 10)R^3 \sin 3\phi$	7th Order 45° Trefoil
Z_{31}	$(56R^6 - 105R^4 + 60R^2 - 10)R^2 \cos 2\phi$	9th Order Astigmatism
Z_{32}	$(56R^6 - 105R^4 + 60R^2 - 10)R^2 \sin 2\phi$	9th Order 45° Astigmatism
Z_{33}	$(126R^8 - 280R^6 + 210R^4 - 60R^2 + 5)R \cos \phi$	9th Order x-Coma
Z_{34}	$(126R^8 - 280R^6 + 210R^4 - 60R^2 + 5)R \sin \phi$	9th Order y-Coma
Z_{35}	$252R^{10} - 630R^8 + 560R^6 - 210R^4 + 30R^2 - 1$	9th Order Spherical
Z_{36}	$924R^{12} - 2772R^{10} + 3150R^8 - 1680R^6 + 420R^4 - 42R^2 + 1$	11th Order Spherical

It is the magnitude of the Zernike coefficients that determines the aberration behavior of a lens. They have units of optical path length relative to the wavelength. (Note that different notational schemes are in use for the Zernike coefficients. Another popular scheme labels the piston term Z_1 rather than Z_0 , so that all the indices differ by 1. Unfortunately, there is no universal standard and naming conventions can vary.)

Zernike polynomials have some important properties that make their use extremely attractive. First, they form a complete set over the unit circle, meaning that any piecewise-continuous function can be fitted exactly within this unit circle with an infinite-order Zernike polynomial. Second, they are *orthogonal* over the unit circle. Mathematically, this means that the product of two different polynomial terms, when integrated over the pupil, will always be zero, whereas the integral of one polynomial term squared will be nonzero. Practically, this means that each polynomial term behaves independently of the others, so that the addition or removal of one polynomial term does not affect the best-fit coefficients of the other polynomial terms. Third, the Zernike polynomial terms are separable into functions of R multiplied by functions of ϕ , providing rotational symmetry. The ϕ functions are trigonometric, so that a rotation of the x -axis (defining ϕ) does not change the form of the equation.

The impact of aberrations on the aerial image can be calculated by modifying the pupil function of the lens given in Chapter 2 to include the phase error due to aberrations.

$$P(f_x, f_y) = P_{\text{ideal}}(f_x, f_y) e^{i2\pi W(R, \phi)} \quad (3.4)$$

Figure 3.6 shows several examples of plots of $W(R, \phi)$ for different simple combinations of third-order Zernike terms. Implicit in the use of Equation (3.4) is that this lens pupil function is also a function of field position and wavelength. The wavelength dependence is discussed in section 3.1.6. While the field dependence of the pupil function is sometimes modeled mathematically (the Seidel aberration equations being the most common), in lithography applications the most common approach is to characterize each field position with its own pupil function.

Although the phase variation across the pupil of a typical lithographic lens can be quite complicated, there are a few simple examples that can be used to describe the majority of the lithographic effects due to aberrations. It is also important to note that the impact of a given aberrated lens on the printing of a mask feature is a strong function of the feature itself. The reason is quite simple: different mask features diffract light differently, sending light through different parts of the lens. As an example, a pattern of equal lines and spaces illuminated with coherent (single direction) light will result in a diffraction pattern of discrete diffraction ‘orders’, rays of light passing through the lens at certain distinct points. Changing the pitch of the line/space pattern changes the position within the lens that the diffraction orders will pass (Figure 3.7). We can think of the diffraction patterning as ‘sampling’ the lens differently depending on the mask pattern. In general, the phase error seen by these different mask patterns will be different, giving rise to feature type, size and orientation dependencies to the effects of aberrations.

3.1.3 Aberration Example – Tilt

One of the simplest forms of an aberration is *tilt*, a linear variation of phase from one side of the lens to the other as given by the Zernike terms Z_1 and Z_2 (Figure 3.8a). As the

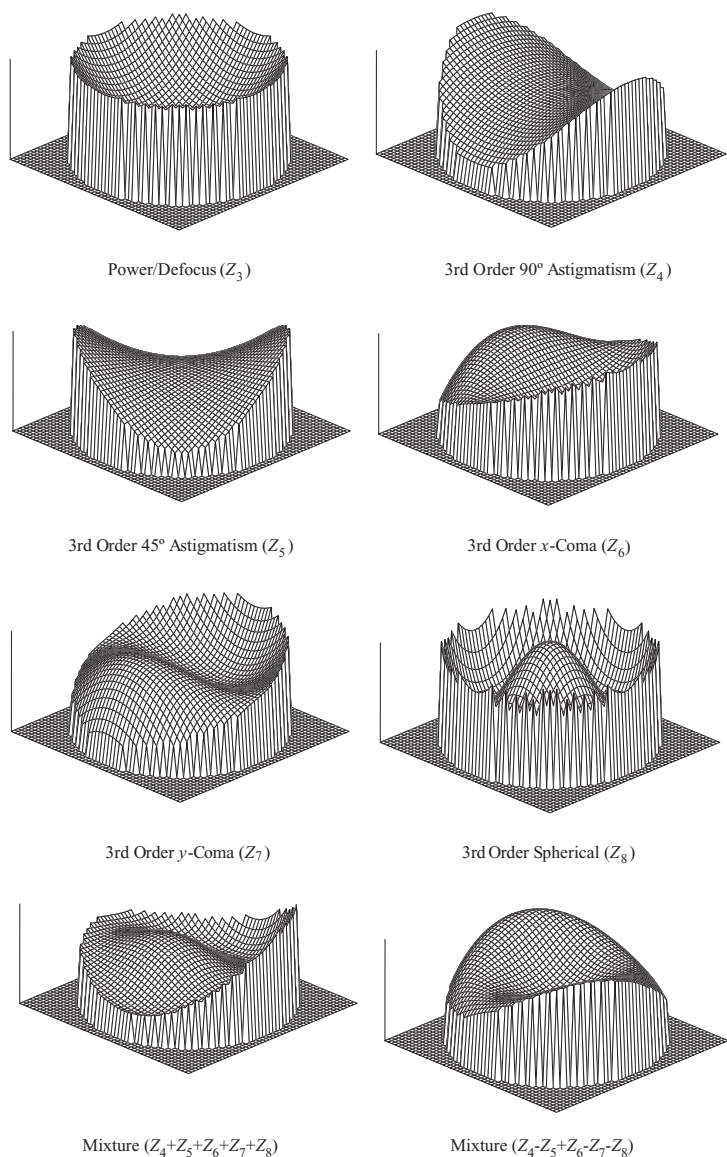


Figure 3.6 Example plots of aberrations (phase error across the pupil)

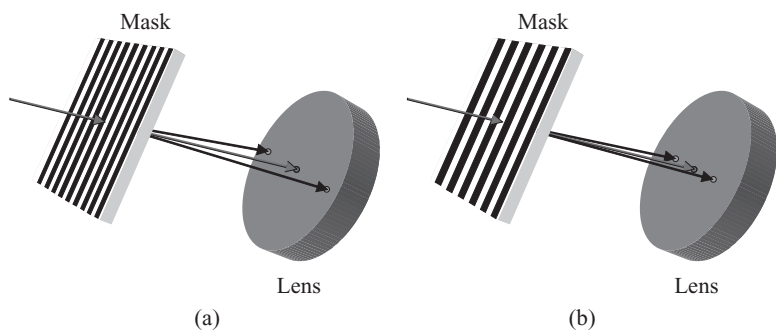


Figure 3.7 Diffraction patterns from (a) a small pitch, and (b) a larger pitch pattern of lines and spaces will result in light passing through a lens at different points in the pupil. Note also that y-oriented line/space features result in a diffraction pattern that samples the lens pupil only along the x-direction

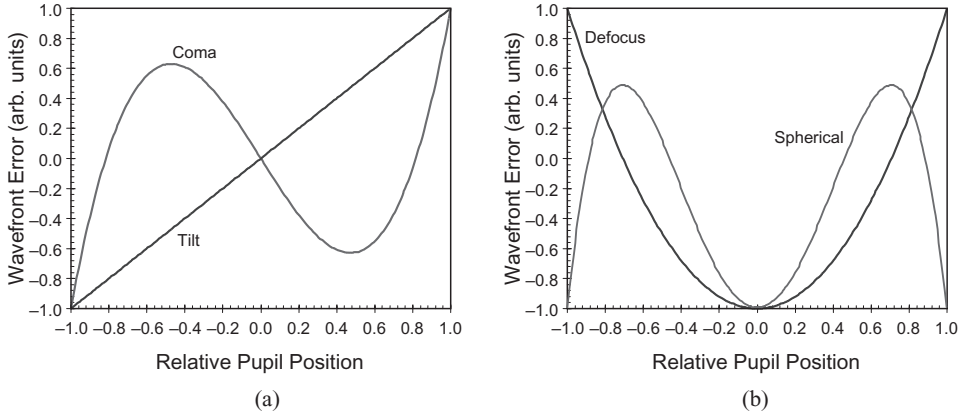


Figure 3.8 Phase error across the diameter of a lens for several simple forms of aberrations: (a) the odd aberrations of tilt and coma, and (b) the even aberrations of defocus and spherical

name implies, the aberration of tilt will ‘tilt’ the wavefront exiting the lens, changing the direction of the light. The result is a simple shift in the position of the final printed feature (as can be easily shown using the shift property of the Fourier transform). For pure tilt, the resulting positional error of the pattern is independent of feature size and type, but is very dependent on the orientation of the pattern. As an example, a pattern of lines and spaces oriented in the y -direction will give rise to a diffraction pattern with diffraction orders spread out in the x -direction (Figure 3.7). The resulting placement error (the amount that the image has shifted in x with respect to the unaberrated image) will be

$$\text{placement error} = -\frac{Z_1 \lambda}{NA} \quad (3.5)$$

In this case, an x -tilted phase variation will cause a pattern placement error for the features, whereas a y -tilt aberration (given by the coefficient Z_2) will have no effect.

The variation of the tilt Zernike coefficients with field position is called *distortion*. Since tilt causes the same image placement error for all features, distortion is independent of feature size and type. As will be seen below in the section on coma, higher-order odd aberrations will cause pattern placement errors as well, but with varying placement error as a function of feature size and type.

3.1.4 Aberration Example – Defocus, Spherical and Astigmatism

The next simplest form of an aberration is called *paraxial defocus* (also called power), a parabolic variation in phase about the center of the lens (Figure 3.8b). Since this aberration is radially symmetric, all feature orientations will behave the same way. However, smaller features, with the diffraction orders farther out toward the edge of the lens, will have a greater amount of phase error. Thus, small features are more sensitive to the

aberration of defocus. More details of the effects of defocus are given in section 3.4. Further explanation of the lithographic impact of focus errors is given in Chapter 8.

There is a whole class of aberrations that behave somewhat like defocus across a given radial cut through the lens pupil. These aberrations, called *even* aberrations, have the same phase error at either side of the center of the lens along any given diagonal (that is, for a fixed ϕ). For even aberrations, the primary effect on a single feature size and orientation is basically a shift in best focus. In general, different feature sizes and orientations will experience a different amount of the even aberration phase error, giving different focal shifts. Spherical aberration can be thought of as causing a focus error that varies with pitch. As Figure 3.8b shows, different pitches, which will give different positions of the first diffraction order, will experience different phase errors and thus different effective focus errors (see Problem 3.5).

A variation in best focus as a function of the orientation of the features is called *astigmatism*. Examining the Zernike terms 4 and 5, for a given orientation of line (and thus a given ϕ), the variation of phase with R is quadratic. Thus, for a given orientation, astigmatism adds a phase error identical to paraxial defocus. But the amount of effective defocus error due to astigmatism will depend on the orientation of the line (that is, on ϕ). One of the major consequences of the presence of astigmatism is a focus-dependent horizontal–vertical (H–V) bias, a difference in linewidth between horizontally oriented and vertically oriented patterns. More on H–V bias will be given in Chapter 8.

3.1.5 Aberration Example – Coma

Finally, *odd* aberrations are those that do not have radial symmetry about the center of the lens. Coma is a common example of an odd aberration (see Figure 3.8). Consider, for example, the sixth term in the Zernike polynomial, representing x -oriented 3rd order coma:

$$Z_6(3R^2 - 2)R \cos \phi \quad (3.6)$$

y -oriented coma (Zernike term number 7) has the same form as the above expression, but with a $\sin \phi$ instead of the cosine term. The combination of these two terms can define any amount of 3rd order coma directed in any orientation. For simplicity, then, we can consider coma in the direction of maximum magnitude as given by the above equation with the angle equal to zero.

What are the main lithographic effects of 3rd order coma? One important effect is a feature-size dependent pattern placement error. Since the x -tilt Zernike term takes the form $Z_1 R \cos \phi$, comparison with Equation (3.6) shows that coma acts as an effective tilt with coefficient $Z_6(3R^2 - 2)$. Consider a pattern of lines and spaces of pitch p . Diffraction will produce discrete orders (directions) of light traveling through the lens, the zero order going straight through the lens and the plus and minus first orders entering the lens at relative radial positions of $R = \pm \lambda/pNA$. For coherent illumination, the pattern placement error can be derived directly from the above expressions for x -tilt and 3rd order x -coma.

$$\text{placement error} = -\frac{Z_6 \lambda}{NA} \left(3 \left(\frac{\lambda}{pNA} \right)^2 - 2 \right) \quad (3.7)$$

It is important to note that the above expression only applies to line/space patterns when $\lambda/NA < p < 3\lambda/NA$ and for coherent illumination. The effect of partially coherent illumination is to spread out the orders in a circle centered about each coherent diffraction order, thus averaging out some of the tilt caused by the coma. Figure 3.9 shows how pattern placement error varies with pitch for a pattern of equal lines and spaces in the presence of coma for different amounts of partial coherence. One can observe a general, though not universal, trend: lower partial coherence increases the sensitivity to aberrations.

A second important effect of coma is asymmetric linewidth errors. A single isolated line, printed in the presence of coma, will show an asymmetry of the right to left side of the photoresist profile. An even greater effect, however, is the difference in linewidth between the rightmost and leftmost lines of a five-bar pattern. Figure 3.10 shows how the difference between right and left linewidths varies almost linearly with the amount of coma, with smaller features more sensitive to a given amount of coma. In fact, this lithographic measurement can be used as a test for the amount of coma in a lens.

The third important impact of coma is a change in the shape of the final resist profile, especially through focus. Coma causes an asymmetric degradation of the resist profile, affecting one side of the feature more than the other, as seen in Figure 3.11. At best focus, this degradation in profile shape is usually not noticeable. But out of focus, there is a characteristic loss of resist off one side of the profile at the part of the profile that is farthest away from the plane of best focus. The reason for this behavior can be seen from aerial image *isophotes* (contours of constant intensity through focus and horizontal position) in the presence of coma, as shown in Figure 3.12.

3.1.6 Chromatic Aberrations

An assumption of the Zernike description of aberrations is that the light being used is monochromatic. The phase error of light transmitted through a lens applies to a specific

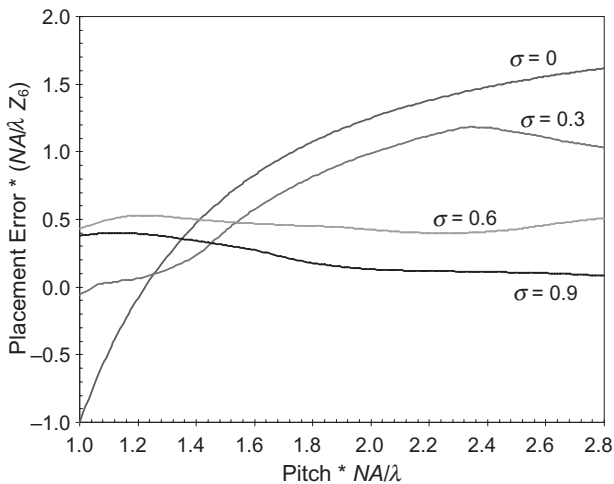


Figure 3.9 The effect of coma on the pattern placement error of a pattern of equal lines and spaces (relative to the magnitude of the 3rd order x-coma Zernike coefficient Z_6) is reduced by the averaging effect of partial coherence

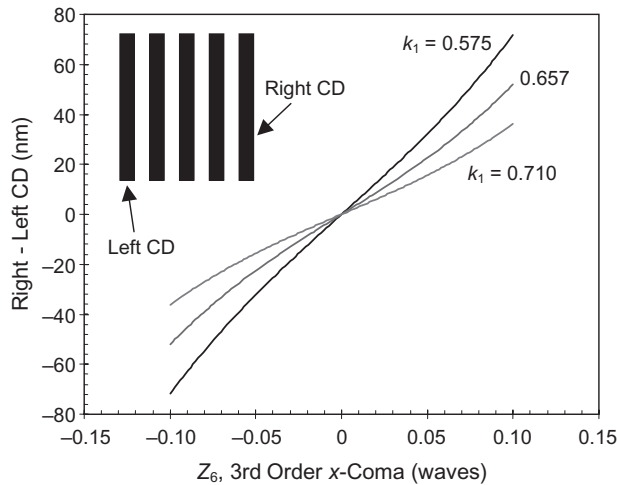


Figure 3.10 The impact of coma on the difference in linewidth between the rightmost and leftmost lines of a five bar pattern (simulated for *i*-line, NA = 0.6, sigma = 0.5). Note that the *y*-oriented lines used here are most affected by *x*-coma. Feature sizes (350, 400 and 450 nm) are expressed as $k_1 = \text{linewidth} * \text{NA} / \lambda$



Figure 3.11 Variation of the resist profile shape through focus in the presence of coma

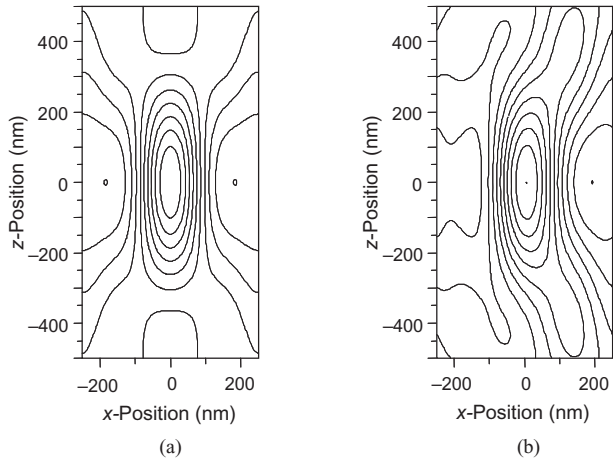


Figure 3.12 Examples of isophotes (contours of constant intensity through focus and horizontal position) for (a) no aberrations, and (b) 100mλ of 3rd order coma. (NA = 0.85, $\lambda = 248\text{ nm}$, $\sigma = 0.5$, 150-nm space on a 500-nm pitch)

wavelength. One expects this phase error, and thus the coefficients of the resulting Zernike polynomial, to vary with the wavelength of the light used. The property of a lens element that allows light to bend is the index of refraction of the lens material. Since the index of refraction of all materials varies with wavelength (a property called *dispersion*), lens elements will focus different wavelengths differently. This fundamental problem, called *chromatic aberration*, can be alleviated by using two different glass materials with different dispersions such that the chromatic aberrations of one lens element cancel the chromatic aberrations of the other. The cemented doublet of Figure 3.2 is the simplest example of a chromatic-corrected lens (called an *achromat*) where each element is made from a glass with different dispersion characteristics. As with all aberrations, this cancellation is not perfect, meaning that all lenses will have some level of residual chromatic aberrations.

The effects of chromatic aberrations depend on two things: the degree to which the Zernike polynomial coefficients vary with wavelength (the magnitude of the chromatic aberrations), and the range of wavelengths used by the imaging tool. For example, a typical i-line stepper might use a range of wavelengths on the order of 5–10 nm, whereas a KrF excimer laser-based deep-UV stepper may illuminate the mask using light with a wavelength range of less than 1 pm. The obvious difference in the magnitude of the wavelength ranges between these two light sources does not mean, however, that chromatic aberrations are a problem for i-line but not for deep-UV. i-line lenses are designed to use the (relatively) wide range of wavelengths by chromatically correcting the lens with several different types of glass. A typical deep-UV lens, on the other hand, makes no attempt at chromatic correction since only fused silica is used for all the elements in the lens (due to a lack of suitable alternative materials). As a result, chromatic aberrations are a concern in deep-UV lithographic lenses even when extremely narrow bandwidth light sources are used.

In principle, every Zernike coefficient is a function of wavelength. In practice, for lenses with no chromatic corrections (i.e. where every lens element is made from the same material), one Zernike term dominates: the term that describes defocus. The plane of best focus shifts with changes in wavelength in a nearly linear fashion, a phenomenon called *longitudinal chromatic aberration*. This can be seen quite readily by considering an ideal thin lens in air. Such a lens is made of glass of index n and has front and back radii of curvature R_1 and R_2 . The focal length f , reduction ratio R , and distances from the lens to the object and image, d_o and d_i , respectively, are all related by the thin-lens equations:

$$\frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) = \frac{1}{d_o} + \frac{1}{d_i}, \quad R = \frac{d_o}{d_i} \quad (3.8)$$

A change in refractive index of the lens (with wavelength, for example) results in a change in focal length. For a fixed object distance and reduction ratio, a change in focal length changes the image distance, that is, focus. Taking the derivative with respect to wavelength,

$$\frac{\Delta focus}{\Delta \lambda} = \frac{d(d_i)}{d\lambda} = -\frac{d_i}{(n-1)} \frac{dn}{d\lambda} \quad (3.9)$$

The term $dn/d\lambda$ is called the dispersion of the lens material. For KrF-grade fused silica at 248.3 nm, $n = 1.508$ and the dispersion is about $-5.6 \times 10^{-7}/\text{pm}$. As an example, a thin lens of fused silica with an image distance of 0.2 m would result in a change in focus per change in wavelength of $0.22 \mu\text{m}/\text{pm}$. Of course, more complicated lens systems would have an effective image distance different from the physical image distance, but the basic trend and magnitude for a lens with no chromatic correction is similar. For immersion lithography, the slope of the focus versus wavelength dependence will increase by the fluid refractive index.

Since the center wavelength of most excimer lasers is easily adjustable over a reasonably large range, this effect can be readily measured for any given stepper. Figure 3.13a shows a typical example.² To account for this behavior, the Zernike term Z_3 (the paraxial defocus term) can be used. The coefficient of the third Zernike term Z_3 can be related to a focus shift $\Delta\delta$ by

$$Z_3 = \Delta\delta \frac{NA^2}{4\lambda_0} \quad (3.10)$$

where λ_0 is the center wavelength of the illumination spectrum. Each wavelength produces a different focus shift $\Delta\delta$, which leads to a different value of Z_3 . Higher-order focus terms could be added as well.

A typical defocus/wavelength behavior for a KrF (248 nm) lens is shown in Figure 3.13a. An excimer laser light source will emit light over a range of wavelengths with a characteristic spectrum (output intensity versus wavelength) that can be roughly approximated as somewhere between a Gaussian shape and a Lorentzian shape. The full width half maximum (FWHM) of the laser output spectrum is called the bandwidth of the laser, with typical line-narrowed KrF and ArF lasers having less than 1-pm bandwidths. Each wavelength in the laser spectrum will be projected through the imaging lens, forming an aerial image shifted in focus according to the wavelength response characteristic for that

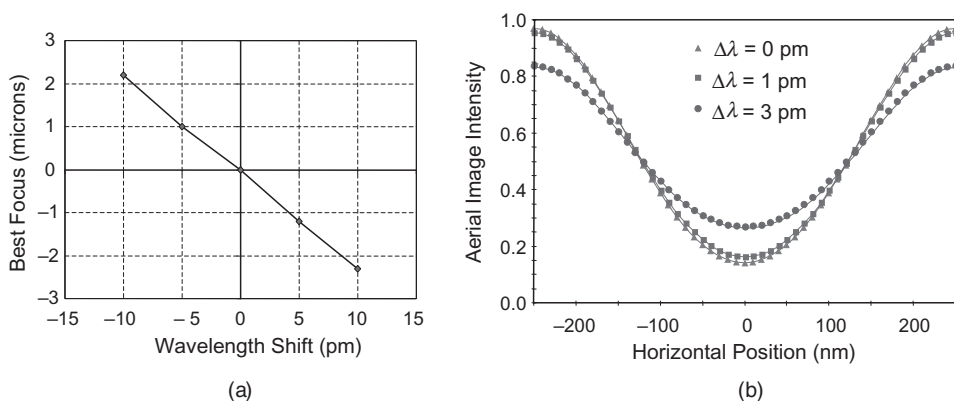


Figure 3.13 Chromatic aberrations: (a) measurement of best focus as a function of center wavelength shows a linear relationship with slope $0.255 \mu\text{m}/\text{pm}$ for this 0.6 NA lens; (b) degradation of the aerial image of a 180-nm line (500-nm pitch) with increasing illumination bandwidth for a chromatic aberration response of $0.255 \mu\text{m}/\text{pm}$

lens. The total aerial image will be the sum of all the images from all the wavelengths in the source, resulting in a final aerial image that is somewhat smeared through focus.

Figure 3.13b shows a typical example. For this 180-nm image, the 1-pm bandwidth shows a slight degradation of the aerial image, but the 3-pm bandwidth is clearly unacceptable. As one might expect, smaller features will be more sensitive to the effects of chromatic aberrations than larger features. Thus, as features are pushed to smaller sizes, even these seemingly miniscule sub-picometer excimer bandwidths will require careful consideration as to their chromatic aberration effects. Also, since dense line/space features tend to respond very differently to focus errors compared to isolated lines, changes in laser bandwidth can result in noticeable changes in *iso-dense bias*, the difference in resist linewidth between an isolated and a dense feature of the same mask width.

For rigorous treatment of chromatic aberrations, a mathematical description of the light source spectral output is required. A modified Lorentzian distribution has been shown to match measured excimer laser spectrums quite well.³ Letting I_{rso} be the relative spectral output intensity,

$$I_{\text{rso}}(\lambda) = \frac{\Gamma^n}{\Gamma^n + [2(\lambda - \lambda_0)]^n} \quad (3.11)$$

where λ_0 is the center wavelength, Γ is the bandwidth, and n is an empirically determined parameter that is typically between 2 and 3. Figure 3.14 shows a typical example from a KrF laser.

One very simple way to avoid the problem of chromatic aberrations is to use mirrors for the imaging elements. Since mirrors reflect light in about the same way for a wide range of wavelengths, near-perfect chromatic correction is possible. However, mirror optical systems pose many other challenges, not the least of which is the extreme difficulty of using multiple focusing mirrors without those mirrors getting in the way of each other.

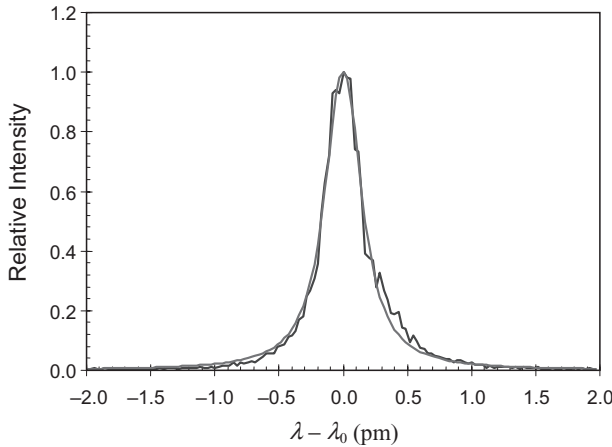


Figure 3.14 Measured KrF laser spectral output¹ and best-fit modified Lorentzian ($\Gamma = 0.34 \text{ pm}$, $n = 2.17$, $\lambda_0 = 248.3271 \text{ nm}$)

3.1.7 Strehl Ratio

The overall level of the aberrations of a lens can be characterized in several ways. The *peak wavefront deviation* is the maximum phase difference (or OPD) between any two points in the pupil (sometimes called the peak-to-valley wavefront error). The *RMS (root mean square) wavefront deviation* provides an average level of phase error or OPD across the pupil. Another approach is to look at the degradation of the point spread function (PSF) caused by the aberrations. Since the PSF samples the imaging lens pupil evenly, its degradation provides an overall measure of how aberrations might affect image quality. Since the ideal PSF is normalized to have a peak value of 1, the *Strehl ratio* is defined as the peak intensity of the PSF of the real lens at best focus, a value that is always less than 1. For anything but the highest levels of aberrations, the Strehl ratio can be related directly to the RMS wavefront deviation:

$$\text{Strehl Ratio} = e^{-(2\pi W_{\text{RMS}})^2} \quad (3.12)$$

where W_{RMS} is the RMS wavefront error in waves. For small levels of aberrations (certainly an applicable constraint for lithographic lenses), the Strehl ratio becomes

$$\text{Strehl Ratio} \approx 1 - (2\pi W_{\text{RMS}})^2 \quad (3.13)$$

As an example, a Strehl ratio of 0.9 corresponds to an RMS wavefront error of 50 milliwaves. For a Strehl ratio of 0.95, the RMS wavefront error is 35 milliwaves. State-of-the-art lithographic lenses will have Strehl ratios greater than 0.95, with the best lenses having RMS wavefront errors of less than 10 milliwaves.

3.2 Pupil Filters and Lens Apodization

Pupil filters are special filters placed inside the objective lens in order to purposely modify the pupil function $P(f_x, f_y)$. In general, the ideal pupil function will provide the best overall imaging capabilities. However, under special circumstances (for example, when imaging one specific mask pattern), a change in the pupil function may result in desirable imaging properties such as enhanced depth of focus. A pupil filter can be described by a pupil filter function $F(f_x, f_y)$ that can have both a variation in transmission (T) and phase (θ) across the pupil:

$$F(f_x, f_y) = T(f_x, f_y) e^{i\theta(f_x, f_y)} \quad (3.14)$$

The final pupil function is then the pupil function given by Equation (3.4) multiplied by the filter function given in Equation (3.14).

In general, lithographic tools used for integrated circuit (IC) production do not allow the insertion of pupil filters. However, the pupil filter function of Equation (3.14) can also be used to describe variations in pupil amplitude transmittance, whether intentionally induced or not, that impact imaging but are not intended for ‘filtering’ purposes. For example, some imaging tools introduce a central obscuration into the pupil in order to extract light for use in through-the-lens alignment systems. Such an obscuration blocks all light transmitted through the central portion of the lens, for example, when the sine of the diffracted angle is less than ± 0.05 . $1\times$ steppers made by Ultratech Stepper and early

steppers made by ASML include such central obscurations for through-the-lens alignment purposes.

Additionally, nonideal lens antireflection coatings can lower the transmittance of the lens at the highest spatial frequencies, sometimes by as much as 5–10%. Such nonideal transmittances can be empirically described by a high-order Gaussian:

$$T(f_x, f_y) = e^{-[(f_x^2 + f_y^2)/NA^2]^q} \quad (3.15)$$

where the power q might be on the order of 5–10. In general, a lowering of the transmittance of a lens at high spatial frequencies is called *apodization*. Such transmission errors can also be described using the Jones pupil, as defined below in section 3.6.4.

3.3 Flare

The goal of an imaging lens is to collect diffracted light, spreading out away from an object, and focus that light down to an imaging plane, creating an image that resembles the original object. A typical lens performs this task through the use of curved surfaces of materials with indices of refraction different than air, relying on the principle of refraction. Implicit in this description is the idea that light travels in only one general direction: from the object to the image. The astute student of optics, however, might detect a problem: the difference in index of refraction between two media (such as air and glass) that gives rise to refraction will also give rise to an unwanted phenomenon – reflection.

The design of a lens, including the ray tracing algorithms used to optimize the individual shapes and sizes of each glass or fused silica element in the lens, makes use of the assumption that all light traveling through the lens continues to travel from object to image *without any reflections*. How is this achieved in practice? One of the hidden technologies of lens manufacturing is the use of antireflection coatings on nearly every glass surface in a lens. These lens coatings are designed to maximize the transmittance of light at the interface between materials through interference (a subject that will be treated extensively in Chapter 4). These coatings, usually made of two layers, have specifically designed thicknesses and refractive indices that reduce reflections through interference among several reflected beams. The coatings are quite effective at reducing reflections at a specific wavelength and over a range of incident angles. They had better be, since a typical microlithographic lens will have over 50 surfaces requiring coatings.

Although the lens coatings used in lithography tools are quite good, they are not perfect. As a result, unwanted reflections, though small in magnitude, are inevitably causing light to bounce around within a lens. Eventually some of this light will make its way to the wafer. For the most part, these spurious reflections are reasonably random, resulting in a nearly uniform background light level exposing the wafer called *flare*. Flare is defined as the fraction of the total light energy reaching the wafer that comes from unwanted reflections and scatterings within the lens.

Lens coating nonideality is not the only source of flare. Flare is caused by anything that causes the light to travel in a ‘nonray trace’ direction. In other words, reflections at an interface, scattering caused by particles or surface roughness, or scattering caused by glass inhomogeneity all result in stray light called flare (Figure 3.15). Defects such as

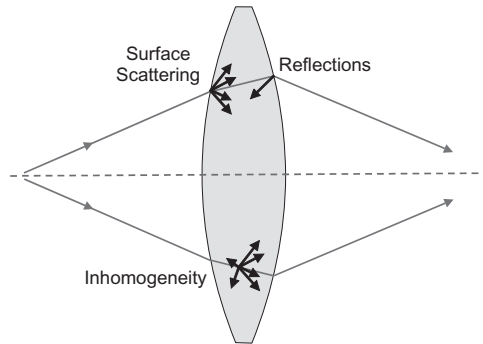


Figure 3.15 Flare is the result of unwanted scattering and reflections as light travels through an optical system

these can be built into the lens during manufacturing, or can arise due to lens degradation (aging, contamination, etc.). External sources of flare, such as light that reflects off the wafer, travels up through the lens and reflects off the backside of the chrome mask, can also contribute. The lithographic consequence of this stray light is probably obvious: degradation of image quality. Since flare is a nearly uniform background light level exposing the wafer, it provides exposure to nominally dark features (such as a line), reducing their quality.

While mirrors do not suffer from the problems of nonideal antireflection coatings inherent in refractive optical elements, they are far more sensitive to surface roughness scattering. For a given RMS surface roughness, a mirror will scatter greater than one order of magnitude more light than an equally rough refractive lens surface.

3.3.1 Measuring Flare

Of course, it is the goal of lens manufacturing and maintenance to produce lenses with very low levels of flare. How can one tell if flare poses a problem for a particular stepper or scanner? Fortunately, measuring flare is reasonably straightforward. Consider the imaging of an island feature whose dimension is extremely large compared to the resolution limits of the imaging tool (say, a 100- μm square island in positive resist). In the absence of flare, the imaging of such a large feature will result in very nearly zero light energy at the center of the image of the island. The presence of flare, on the other hand, will provide light to this otherwise dark region on the wafer (Figure 3.16). A positive photoresist can be used as a very sensitive detector for low levels of flare.

The dose-to-clear (E_0) is defined as the minimum dose required to completely remove the photoresist during development for a large open-frame exposure. A related concept is the island dose-to-clear ($E_{0\text{-island}}$), the minimum dose required to completely wash away a large island structure during a normal development process. In the absence of flare, a large island would take nearly an infinite dose to produce enough exposure at its center to make the resist soluble in developer. With flare present, however, this dose is reduced considerably. In fact, by measuring the normal dose-to-clear and the dose-to-clear for the large island, the amount of flare can be determined as⁴

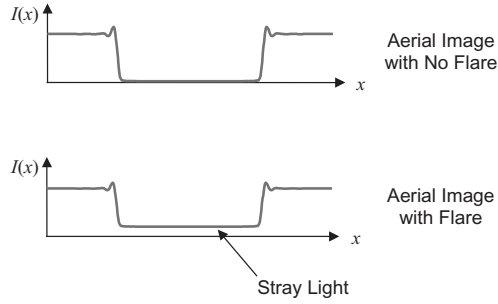


Figure 3.16 Plots of the aerial image intensity $I(x)$ for a large island mask pattern with and without flare

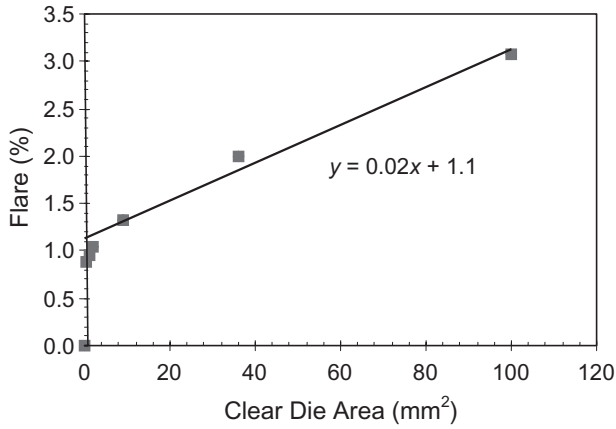


Figure 3.17 Using framing blades to change the field size (and thus total clear area of the reticle), flare was measured at the center of the field (from Mack and Kaufman⁴)

$$Flare = \frac{E_0}{E_{0-\text{island}}} \quad (3.16)$$

For example, if the dose-to-clear of a resist is 35 mJ/cm^2 , then an imaging tool with 5% flare would mean that a large island will clear with a dose of 700 mJ/cm^2 .

Although flare is a characteristic of an imaging tool, it is also a function of how that tool is used. For example, the amount of flare experienced by any given feature is a function of both the local environment around that feature (short range flare) and the total amount of energy going through the lens (long range flare).⁵ A dark field reticle produces images with almost no flare, whereas a reticle which is almost 100% clear will result in the maximum possible flare. The data in Figure 3.17 shows the two distinct regions clearly. Here, a clear field reticle was used with one flare target placed in the middle of the field. Framing blades were used to change the field size and thus the total clear area of the exposure field. As can be seen, flare quickly rises to a level of about 1%, then

grows approximately linearly with clear area of the field. The y-intercept of the linear portion of the curve can be thought of as the short range contribution to the total flare.

Flare is also a function of field position, with points in the center of the field often experiencing flare levels 50% higher than points near the edge of the field. This phenomenon may be thought of as a side effect of the long-range versus short-range scattering discussed above. At the edge of the field, the local clear area is half of that at the center of the field. Finally, flare is also increased when a reflective wafer is exposed. Light reflected back up into the optical system can scatter and find its way back to the wafer as stray light.

3.3.2 Modeling Flare

Consider a simple mechanism for the generation of DC (long range) flare. Light passing through the lens is scattered by one or more of the mechanisms shown in Figure 3.15. In its simplest manifestation, flare would cause all light to scatter in equal proportion regardless of its spatial frequency and field position (angle and position with which it enters the lens). In this case, scattering will result in a uniform reduction in the energy that coherently interacts to form the image. If SF represents the *scatter fraction*, the fraction of the light energy (or intensity) that scatters and thus does not contribute to the coherent creation of the image, the resulting image would be $(1-SF)I_0(x,y)$ where $I_0(x,y)$ is the image that would result if there were no scattering. But where does this scattered light go? Another simple approximation for uniform DC flare would be that the scattered light is uniformly spread over the exposure field. Thus, a DC or background dose would be added to the aerial image. But while scattering reduces the intensity of the image *locally*, the background dose is added *globally*. Thus, the additional background dose would be equal to the scatter fraction multiplied by the total energy passing through the lens:

$$I(x,y) = SF \cdot EF + (1 - SF)I_0(x,y) \quad (3.17)$$

where EF is the energy fraction of the reticle, the ratio of the total energy reaching the wafer for this reticle compared to the energy that would reach the wafer for a perfectly clear reticle. As a reasonable approximation, the total energy passing through the lens is proportional to the clear area of the reticle. (Because of diffraction, only a portion of the energy passing through the reticle actually passes through the lens. For example, for a mask pattern of equal lines and spaces where only the zero and ± 1 first orders pass through the lens, the energy going through the lens is about 90% of the energy that passes through the mask.) Thus, a simple but still quite accurate DC flare model would be

$$I(x,y) = SF \cdot CF + (1 - SF)I_0(x,y) \quad (3.18)$$

where CF is the clear area fraction for the field.

For a 100% clear area full-field mask, the standard measurement of flare for a clear field reticle would yield the scatter fraction used in Equation (3.18). Since the clear area fraction is generally known (at least approximately) for any mask, the DC flare can be calculated for any mask once the scatter fraction for the lens has been measured. For a dark field mask, as the clear area fraction goes to zero, the effect of flare is just a loss of dose for the image. It is doubtful that this dose loss is observable in normal

lithographic practice since dark field exposure layers (such as contacts and vias) generally operate at different numerical aperture/partial coherence combinations than clear field reticles, and dose calibration is not consistent across different stepper lens and illumination settings.

A slight correction to Equation (3.18) would occur if some fraction of the scattered light was assumed to be lost (absorbed into the lens housing, for example). However, for the low levels of flare expected for normal lithographic imaging tools (much less than 10%), the effect of some of the scatter fraction being lost and not reaching the wafer would be a simple dose recalibration and would not be observable.

Equation (3.18) does not take into account the impact of short-range flare effects, as observed in Figure 3.17. The scattering PSF approach⁶ can account for these short-range flare effects. The local scattered light is described as the convolution of the local light intensity (the aerial image) with a scattering point spread function, PSF_{scat} .

$$I_{\text{scat}}(x,y) = PSF_{\text{scat}}(x,y) * I_0(x,y) \quad (3.19)$$

Note that if the scattering PSF is a constant, then the convolution becomes a calculation of the total energy of the aerial image and the scattered intensity is equivalent to the scatter fraction times the energy fraction. Generally this global scattering is removed from the PSF_{scat} and treated separately. A typical scattering PSF is a Gaussian, with standard deviations in the range of 10–20 μm , though other functional forms are used as well.

3.4 Defocus

The inverse Fourier transform expression for calculating the aerial image (see Chapter 2) applies only to the image at the focal plane. What happens when the imaging system is out of focus, i.e. what is the image intensity distribution some small distance away from the plane of best focus?

3.4.1 Defocus as an Aberration

The impact of focus errors on the resulting aerial image can be described as an aberration of a sort. Consider a perfect spherical wave converging (i.e. focusing) down to a point. An ideal projection system will create such a wave coming out of the lens aperture (called the *exit pupil*), as shown in Figure 3.18a. If the wafer to be printed were placed in the same plane as the focal point of this wave, we would say that the wafer was in focus. What happens if the wafer were removed from this plane by some distance δ , called the defocus distance? Figure 3.18b shows such a situation. The spherical wave with the solid line represents the actual wave focused to a point a distance δ away from the wafer (this defocus distance is arbitrarily defined to be positive when the wafer is moved away from the lens). If, however, the wave had a different shape, as given by the dotted curve, then the wafer would be in focus. Note that the only difference between these two different waves is the radius of curvature of the wavefront. Since the dotted curve is the wavefront we want for the given wafer position, we can say that the actual wavefront is in error because it does not focus where the wafer is located. (This is just a variation of ‘the

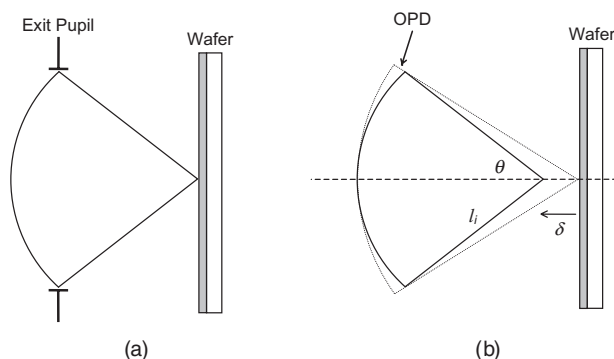


Figure 3.18 Focusing of light can be thought of as a converging spherical wave: (a) in focus, and (b) out of focus by a distance δ . The optical path difference (OPD) can be related to the defocus distance δ , the angle θ and the radius of curvature of the converging wave (also called the image distance) l_i

customer is always right' attitude – the wafer is always right; it is the optical wavefront that is out of focus.)

By viewing the actual wavefront as having an error in curvature relative to the desired wavefront (i.e. the one that focuses on the wafer), we can quantify the effect of defocus. Looking at Figure 3.18b, it is apparent that the distance from the desired to the 'defocused' wavefront goes from zero at the center of the exit pupil and increases as we approach the edge of the pupil. This distance between wavefronts is called the *optical path difference* (OPD). The OPD is a function of the defocus distance and the position within the pupil and can be obtained from the geometry shown in Figure 3.18b. Describing the position within the exit pupil by the light propagation angle θ , the OPD (assuming $l_i \gg \delta$) is given by

$$OPD = n\delta(1 - \cos\theta) \quad (3.20)$$

where n is the refractive index of the medium between the lens and the wafer (generally air, though possibly an immersion fluid, as discussed in section 3.7).

As we have seen before, the spatial frequency and the NA define positions within the pupil as the sine of an angle. Thus, the above expression for OPD will be more useful if expressed as a function of $\sin\theta$:

$$OPD = n\delta(1 - \cos\theta) = n\delta(1 - \sqrt{1 - \sin^2\theta}) \quad (3.21)$$

Expanding the square root as a Taylor series,

$$OPD = \frac{1}{2}n\delta\left(\sin^2\theta + \frac{\sin^4\theta}{4} + \frac{\sin^6\theta}{8} + \dots\right) \approx \frac{1}{2}n\delta\sin^2\theta \quad (3.22)$$

where the final approximation is accurate only for relatively small angles. Figure 3.19 shows the accuracy of this approximation as a function of the angle θ .

So how does this OPD affect the formation of an image? The OPD acts just like an aberration, modifying the pupil function of the lens. For light, this path length traveled

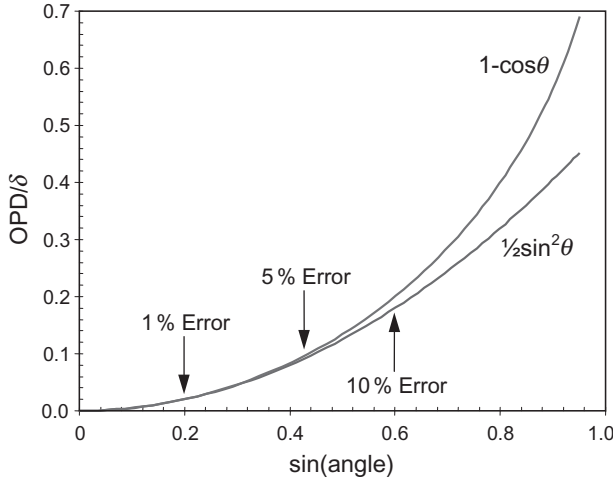


Figure 3.19 Comparison of the exact and approximate expressions for the defocus optical path difference (OPD) shows an increasing error as the angle increases. An angle of 37° (corresponding to the edge of an $\text{NA} = 0.6$ lens) shows an error of 10 % for the approximate expression. At an NA of 0.93, the error in the approximate expression is 32 %

(the OPD) is equivalent to a change in phase. Thus, the OPD can be expressed as a phase error, $\Delta\Phi$, due to defocus:

$$P_{\text{defocus}}(f_x, f_y) = P_{\text{ideal}}(f_x, f_y) e^{i2\pi \text{OPD}/\lambda} = P_{\text{ideal}}(f_x, f_y) e^{i\Delta\Phi} \quad (3.23)$$

$$\Delta\Phi = 2\pi \text{OPD}/\lambda = 2\pi n\delta(1 - \cos\theta)/\lambda \approx \pi n\delta \sin^2\theta/\lambda$$

where again, the final approximation is only valid for small angles. We are now ready to see how defocus affects the diffraction pattern and the resulting image. Our interpretation of defocus is that it causes a phase error as a function of radial position within the aperture. Light in the center of the aperture has no error; light at the edge of the aperture has the greatest phase error. This is very important when we remember what a diffraction pattern looks like as it enters the lens aperture. Recall that diffraction by periodic patterns results in discrete diffraction orders – the zero order is the undiffracted light passing through the center of the lens; higher orders contain information necessary to reconstruct the image. Thus, the effect of defocus is to add a phase error to the higher-order diffracted light relative to the zero order. When the lens recombines these orders to form an image, this phase error will result in a degraded image (Figure 3.20).

Considering defocus as an aberration, the pupil error due to defocus [Equation (3.21)] can be expressed in the terms of a Zernike polynomial. In the paraxial approximation (small angles), there is a simple correspondence to the third term in the polynomial expression (3.2).

$$Z_3 \approx \frac{\delta \text{NA}^2}{4\lambda} \quad (3.24)$$

If the exact defocus expression is used, higher-order Zernike defocus terms are required (Z_8 , Z_{15} , Z_{24} , etc.).

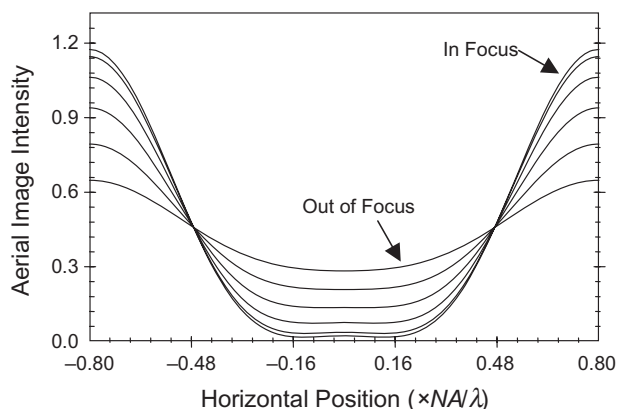


Figure 3.20 Aerial image intensity of a $0.8\lambda/\text{NA}$ line and space pattern as focus is changed

3.4.2 Defocus Example: Dense Lines and Spaces and Three-Beam Imaging

As an example of using Equation (3.23) when calculating an aerial image, consider a pattern of small lines and spaces so that only the 0 and ± 1 diffraction orders are used to form the image (that is, three-beam imaging). For coherent illumination, the zero order will pass through the center of the lens and undergo no phase error due to defocus. The first diffracted orders will pass through the pupil at spatial frequencies of $\pm 1/p$ and experience a phase error of

$$\Delta\Phi = 2\pi n\delta(1 - \sqrt{1 - (\lambda/np)^2})/\lambda \quad (3.25)$$

Calculating the electric field image including the pupil function defocus,

$$E(x) = \int_{-\infty}^{\infty} \sum_{j=-1}^1 a_j \delta\left(f_x - \frac{j}{p}\right) e^{i\Delta\Phi} e^{i2\pi f_x x} df_x \quad (3.26)$$

$$E(x) = a_0 + 2a_1 e^{i\Delta\Phi} \cos(2\pi x/p)$$

The intensity image becomes

$$I(x) = a_0^2 + 4a_0a_1 \cos(\Delta\Phi) \cos(2\pi x/p) + 2a_1^2 [1 + \cos(4\pi x/p)] \quad (3.27)$$

The impact of defocus is to reduce the magnitude of the main cosine term in the image. In other words, as the first orders go out of phase with respect to the zero order, the interference between the zero and first orders decreases. When $\Delta\Phi = 90^\circ$, there is no interference and the $\cos(2\pi x/p)$ term disappears.

Partial coherence can significantly change the focus response of an aerial image. The effect of partial coherence is to spread the diffraction order points into larger spots, the shape of each order's spot being determined by the shape of the source. Each point on the source is independent, i.e. incoherent, with no fixed phase relationship to any other source point. Each source point produces a coherent aerial image, the total image

being the (incoherent) sum of all the intensity images from each source point. Since only the exact center source point produces an image affected by focus in the way described by Equation (3.27), a more thorough treatment is required for partially coherent imaging.

Consider the case, as we did above, where only the first diffraction orders make it through the lens and that these orders are completely inside the lens (not clipped by the aperture). This occurs when

$$\frac{\lambda}{NA(1-\sigma)} < p < \frac{3\lambda}{NA(1+\sigma)} \quad (3.28)$$

A quick look at this constraint shows that it can possibly be true only when $\sigma < 0.5$. Thus, while our example here will not be applicable to many situations, it will still be informative. Second, we shall use the paraxial approximation for the effects of focus, so that the OPD due to a defocus error δ will be a quadratic function of the sine of the incident angle or the spatial frequency.

Now the aerial image in the presence of defocus can be calculated analytically by integrating over the source. Considering only equal lines and spaces for the sake of simplicity,

$$I(x) = a_0^2 + 4a_0a_1 \cos(\Delta\Phi) \frac{2J_1(a)}{a} \cos(2\pi x/p) + 2a_1^2 \left[1 + \frac{J_1(2a)}{a} \cos(4\pi x/p) \right] \quad (3.29)$$

where $a = 2\pi\delta\sigma NA/p$ and J_1 is the Bessel function of the first kind, order one. This defocus function $2J_1(a)/a$ is plotted in Figure 3.21 and has the familiar Airy disk form. Comparing Equations (3.27) and (3.29), the effect of partial coherence is to cause a faster decline in the main cosine interference term [the Airy disk falloff is on top of the $\cos(\Delta\Phi)$ falloff

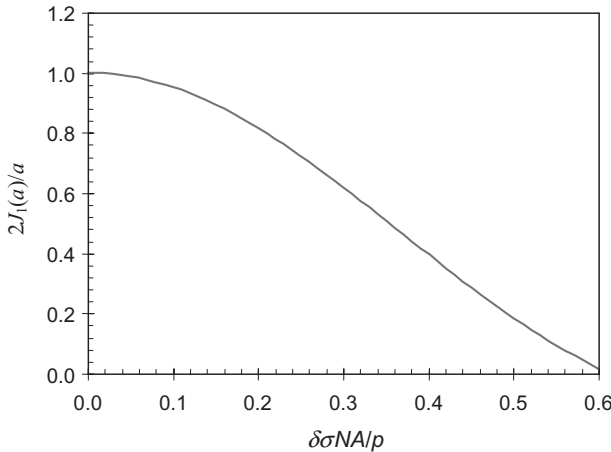


Figure 3.21 The Airy disk function as it falls off with defocus

that occurs for coherent illumination] and to add a degradation to the higher-order interference of the + first and – first orders.

It can be useful to simplify the above aerial image Equation (3.29) for the case of small amounts of defocus. Both the cosine term and the Airy disk terms can be expanded as series approximations, keeping only the low-order terms:

$$\begin{aligned}\cos(\Delta\Phi) &\approx 1 - \frac{\Delta\Phi^2}{2} \\ \frac{2J_1(a)}{a} &\approx 1 - \frac{a^2}{8}\end{aligned}\tag{3.30}$$

Using these approximations, and keeping only terms out to second order in defocus,

$$\begin{aligned}I(x, \delta) &\approx I(x, 0) - 2a_0a_1 \cos(2\pi x/p) \left(\Delta\Phi^2 + \frac{a^2}{4} \right) - a_1^2 \cos(4\pi x/p) a^2 \\ &= I(x, 0) - \left(\frac{\pi\delta\lambda}{p^2} \right)^2 \left[2a_0a_1 \cos(2\pi x/p) \left(1 + \left(\frac{\sigma NAp}{\lambda} \right)^2 \right) - \right. \\ &\quad \left. 4a_1^2 \cos(4\pi x/p) \left(\frac{\sigma NAp}{\lambda} \right)^2 \right]\end{aligned}\tag{3.31}$$

Thus, the final result is an example where the effects of defocus can be put into a separable form as

$$I(x, \delta) \approx I(x, 0) - \delta^2 f(x)\tag{3.32}$$

Most images can be put into such a form for small amounts of defocus.

3.4.3 Defocus Example: Dense Lines and Spaces and Two-Beam Imaging

Consider the case of coherent off-axis illumination of a dense line/space pattern. In particular, let's assume that the illumination is a plane wave tilted such that the diffraction pattern is shifted by f'_x and only the zero and –1st diffraction orders pass through the lens. The portion of the diffraction pattern passing through the lens will be

$$T_m(f_x)P(f_x) = a_0\delta(f_x - f'_x) + a_1\delta\left(f_x + \frac{1}{p} - f'_x\right)\tag{3.33}$$

With defocus included in the pupil function, the electric field image becomes

$$\begin{aligned}E(x) &= a_0 e^{i\Delta\Phi_0} e^{i2\pi f'_x x} + a_1 e^{i\Delta\Phi_1} e^{-i2\pi x/p} e^{i2\pi f'_x x} \\ &= e^{i\Delta\Phi_0} e^{i2\pi f'_x x} \left[a_0 + a_1 e^{i(\Delta\Phi_1 - \Delta\Phi_0)} e^{-i2\pi x/p} \right]\end{aligned}\tag{3.34}$$

where the phase errors for the zero and first orders are, respectively,

$$\Delta\Phi_0 = 2\pi n\delta \left(1 - \sqrt{1 - (\lambda f'_x)^2} \right) / \lambda, \quad \Delta\Phi_1 = 2\pi n\delta \left(1 - \sqrt{1 - \left(\lambda \left[\frac{1}{p} - f'_x \right] \right)^2} \right) / \lambda\tag{3.35}$$

The intensity image is

$$I(x) = \left| a_0 + a_1 e^{i(\Delta\Phi_1 - \Delta\Phi_0)} e^{-i2\pi x/p} \right|^2 = a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p - \Delta\Phi_1 + \Delta\Phi_0) \quad (3.36)$$

Note that the final image depends only on the difference in phase between the zero and first orders.

The impact of defocus on this tilted-illumination image is clear from Equation (3.36): focus errors cause a shift in the position of the sinusoidal image. Such a focus-dependent pattern placement error is called *nontelecentricity*. *Telecentricity*, on the other hand, means that the resulting image will not shift in position as the wafer goes out of focus. Wafer-side telecentricity is a requirement for lithographic imaging. Telecentricity can be recovered by adding a second illumination angle, causing the same shift in the diffraction pattern but in the opposite direction (that is, a shift of $-f'_x$). The resulting image will be the same as that given by Equation (3.36), except with the opposite sign of the phase error. The average of the two aerial images coming from these two symmetrically tilted plane waves will be

$$I(x) = \frac{I(x, f'_x) + I(x, -f'_x)}{2} = a_0^2 + a_1^2 + 2a_0a_1 \cos(\Delta\Phi_1 - \Delta\Phi_0) \cos(2\pi x/p) \quad (3.37)$$

Consider now a very special case, where the tilt of the illumination is chosen to be $f'_x = \pm 1/2p$. For this case, the zero order and the first order will be evenly spaced about the center of the lens and the phase error due to defocus will be the same for both orders. Thus, the resulting intensity image is

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1 \cos(2\pi x/p) \quad (3.38)$$

Remarkably, this is the same image that is obtained when at best focus. That is, defocus has no impact on the final aerial image for this special case of coherent two-beam imaging where the zero and first orders are evenly spaced about the center of the lens. The possibility of essentially infinite depth of focus when the tilt is adjusted properly for a specific pitch is the reason why off-axis illumination is so popular for high-resolution imaging.

In reality, the infinite depth of focus promised by Equation (3.38) is never realized because real illumination sources are not coherent – they are partially coherent. Thus, instead of two point sources separated by $\pm f'_x$, a real source would be two disks centered at these spatial frequencies. The resulting illumination is called dipole illumination. The image from such a dipole source, assuming that all of the zero order and one first order go through the lens, would be the integral of Equation (3.37) over the range of spatial frequency shifts corresponding to the full shape of the source. Using the paraxial approximation for the effects of focus, so that the OPD due to a defocus error δ will be a quadratic function of the sine of the incident angle or the spatial frequency, the phase difference from zero to first orders for any given source point is

$$\Delta\Phi_1 - \Delta\Phi_0 \approx \pi n \delta \lambda \left[\left(\frac{1}{p} - f'_x \right)^2 - f_x'^2 \right] = \frac{\pi n \delta \lambda}{p} \left[\frac{1}{p} - 2f'_x \right] \quad (3.39)$$

Since each dipole disk is centered at $f'_x = \pm 1/2 p$, let a point on this source be expressed as $f'_x = 1/2 p - \Delta f'_x$:

$$\Delta\Phi_1 - \Delta\Phi_0 = \frac{2\pi n\delta\lambda}{p} \Delta f'_x = \frac{2\pi n\delta NA}{p} \Delta\sigma_x \quad (3.40)$$

where the final expression comes from converting spatial frequencies into sigma coordinates.

Using Equation (3.40) in Equation (3.37) and integrating over the source, where each disk has a radius of σ , the resulting two-beam image through focus becomes

$$I(x) = a_0^2 + a_1^2 + 2a_0a_1 \frac{2J_1(a)}{a} \cos(2\pi x/p) \quad (3.41)$$

where $a = 2\pi\delta\sigma NA/p$ and J_1 is the Bessel function of the first kind, order one. Referring again to Figure 3.21, one can see that when the size of the dipole disk is bigger than zero, defocus will cause a degradation in the quality of the aerial image. Note from the definition of the term a that the sensitivity to defocus is in direct proportion to σ , the radius of the dipole disk.

3.4.4 Image Isofocal Point

Figure 3.20 shows an interesting phenomenon where at a particular horizontal position, the aerial image curves cross. In other words, at some x -position for this 1D image, the intensity is not a function of focus (although in general this is only approximately true). The point at which intensity does not depend of focus is called the *image isofocal point*. Mathematically, it is the value of x at which

$$\frac{\partial I}{\partial \delta} = 0 \quad (3.42)$$

Consider the coherent three-beam image case of Equation (3.27). The isofocal point is given by

$$\frac{\partial I}{\partial \delta} = -4a_0a_1 \cos(2\pi x/p) \sin(\Delta\Phi) \left(2\pi n \left(1 - \sqrt{1 - (\lambda/p)^2} \right) / \lambda \right) = 0 \quad (3.43)$$

Solving for x , the isofocal point is $\pm jp/4$, where j is any odd number. Examining the partially coherent dipole two-beam image of Equation (3.41), the same isofocal point is obtained. In fact, this same isofocal point always results whenever the image is made up of 2-beam or a combination of 1-beam and 2-beam imaging.

For the partially coherent three-beam image case, the result is far more complicated. However, a reasonably simple answer comes from looking for the isofocal point for small amounts of defocus, so that Equation (3.31) can be used. Setting the term in this equation that multiplies the defocus to zero,

$$2a_0a_1 \cos(2\pi x/p) \left(1 + \left(\frac{\sigma NA p}{\lambda} \right)^2 \right) = -4a_1^2 \cos(4\pi x/p) \left(\frac{\sigma NA p}{\lambda} \right)^2 \quad (3.44)$$

$$\frac{\cos(4\pi x/p)}{\cos(2\pi x/p)} = -\frac{a_0}{2a_1} \left(1 + \left(\frac{\lambda}{\sigma NA p} \right)^2 \right) \quad (3.45)$$

Consider the case of equal lines and spaces, so that the isofocal value for x can be defined as $p/4 + \Delta x$. Assuming that Δx will be small,

$$\frac{\cos(4\pi x/p)}{\cos(2\pi x/p)} = \frac{\cos(4\pi \Delta x/p)}{\sin(2\pi \Delta x/p)} \approx \frac{p}{2\pi \Delta x} \quad (3.46)$$

Thus, the isofocal point will be approximately at

$$\Delta x \approx -\frac{2p}{\pi^2} \left(1 + \left(\frac{\lambda}{\sigma N A p} \right)^2 \right)^{-1} \quad (3.47)$$

Recall that this approximate position of the image isofocal point applies to three-beam imaging of equal lines and spaces under conventional illumination such that the pupil does not clip any of the first orders. The minus sign shows that the image isofocal point will occur in the space region of the image (since $x = 0$ is the center of the space).

3.4.5 Focus Averaging

Frequently, the final aerial image is an average of the aerial image through some range of focus, a phenomenon called *focus averaging* (also called focus drilling, and sometimes known by the confusing term ‘focus blur’). Chromatic aberrations, discussed in section 3.1.6 produce just such a result, as do stage scanning errors, to be discussed in section 3.5 below. Occasionally, such focus averaging is intentionally induced in order to extend the tolerable range of focus.⁷

The impact of focus averaging can be seen in an approximate form using the separable defocus image expression of Equation (3.32), which is appropriate for small levels of defocus. If $P(\delta)$ is the probability distribution of a defocus error δ , the focus-averaged image will be

$$I(x, \delta) \approx I(x, 0) - f(x) \int_{-\infty}^{\infty} \delta^2 P(\delta) d\delta \quad (3.48)$$

Consider a uniform probability distribution over a range of focus from $\delta_{\min}$ to $\delta_{\max}$. The resulting image will be

$$I(x) \approx I(x, 0) - f(x) \left[\frac{\delta_{\max}^3 - \delta_{\min}^3}{3(\delta_{\max} - \delta_{\min})} \right] \quad (3.49)$$

If the focus averaging range is symmetric about best focus (so that $\delta_{\min} = -\delta_{\max}$),

$$I(x) \approx I(x, 0) - f(x) \left[\frac{\delta_{\max}^2}{3} \right] \quad (3.50)$$

If now some unintentional focus error δ is added,

$$I(x, \delta) \approx I(x, 0) - f(x) \left[\frac{\delta_{\max}^2}{3} + \delta^2 \right] \quad (3.51)$$

Another common probability distribution for focus averaging is a Gaussian with standard deviation σ_F . The resulting image, including unintentional focus error δ , is

$$I(x, \delta) \approx I(x, 0) - f(x)[\sigma_F^2 + \delta^2] \quad (3.52)$$

For an arbitrary focus averaging function, an effective defocus error (assuming small focus errors) can be defined by

$$\delta_{\text{eff}}^2 \approx \sigma_F^2 + \delta^2 \quad (3.53)$$

where σ_F^2 is the variance of the focus averaging probability distribution.

3.4.6 Reticle Defocus

While the main focus problem in optical lithography may be controlling the focal position of the wafer, it is certainly possible for the reticle to be out of focus as well. Like the wafer, reticle focus errors can be described as an OPD with essentially the same form as for wafer focus errors:

$$OPD = n_m \delta_m (1 - \cos \theta_m) \approx \frac{1}{2} n_m \delta_m \sin^2 \theta_m \quad (3.54)$$

where the subscript ‘m’ refers to the mask side. Because of the reduction ratio, angles on the mask side are much lower than on the wafer side, and so the paraxial approximation for the OPD on the right-hand side of Equation (3.54) will be quite accurate. As discussed in Chapter 2, angles on the mask side can be related to angles on the wafer side by

$$R n_m \sin \theta_m = n_w \sin \theta_w \quad (3.55)$$

Thus, the OPD caused by the mask defocus can be described in terms of the wafer-side angles:

$$OPD \approx \frac{1}{2} \left(\frac{\delta_m}{R^2} \right) \left(\frac{n_w}{n_m} \right) n_w \sin^2 \theta_w \quad (3.56)$$

Comparing this expression to the paraxial effect of wafer defocus, the equivalent wafer defocus to cause this same amount of OPD would be

$$\delta_w \approx \left(\frac{\delta_m}{R^2} \right) \left(\frac{n_w}{n_m} \right) \quad (3.57)$$

For imaging in air, where both the wafer-side and mask-side refractive indices are 1, focus errors of the reticle are like focus errors on the wafer, but reduced by the reduction ratio squared. For a typical reduction ratio of 4, this means that the image is 16 times less sensitive to reticle focus errors than to wafer focus errors. (We often say that the longitudinal magnification is equal to the lateral magnification squared.) Of course, this relationship is only approximate since wafer focus errors are not accurately described by the paraxial form of the OPD for reasonably large numerical apertures. However, reticle focus errors can be readily accounted for using the Zernike third-order defocus term:

$$Z_3 = \left(\frac{\delta_m}{R^2} \right) \frac{NA^2}{4\lambda} \quad (3.58)$$

3.4.7 Rayleigh Depth of Focus

Depth of Focus (DOF) is defined as the range of focus that can be tolerated, before the image quality is degraded beyond usefulness. Lord Rayleigh, more than 100 years ago, gave us a simple approach to estimate depth of focus in an imaging system. Here we'll express his method and results in modern lithographic terms. We'll consider the coherent imaging of an array of small lines and spaces so that only the zero and the plus and minus first diffraction orders pass through the lens to form the image.

As discussed above in section 3.4.1, a common way of thinking about the effect of defocus on an image is to consider the defocusing of a wafer as equivalent to causing an aberration – an error in curvature of the actual wavefront relative to the desired wavefront (i.e. the one that focuses on the wafer). Equations (3.20)–(3.22) show how the OPD varies across the pupil for a given amount of defocus. How much OPD can our line/space pattern tolerate? Consider the extreme case. If the OPD of the first diffraction orders were set to a quarter of the wavelength, the zero and first diffracted orders would be exactly 90° out of phase with each other. At this much OPD, the zero order would not interfere with the first orders at all and no pattern would be formed. The true amount of tolerable OPD must be less than this amount:

$$OPD_{\max} = k_2 \frac{\lambda}{4}, \quad \text{where } k_2 < 1 \quad (3.59)$$

Substituting this maximum permissible OPD into Equation (3.20), we can find the DOF.

$$DOF = 2\delta_{\max} = \frac{k_2}{2} \frac{\lambda}{n(1 - \cos \theta)} \quad (3.60)$$

At the time of Lord Rayleigh, lens numerical apertures were relatively small and the medium of imaging was air, so that $n = 1$. Thus, the largest angles going through the lens were also quite small and the cosine term could be approximated in terms of $\sin \theta$:

$$DOF \approx k_2 \frac{\lambda}{\sin^2 \theta} = k_2 \frac{p^2}{\lambda} \quad (3.61)$$

This paraxial form is reasonably accurate for numerical apertures below 0.5. At this point, Lord Rayleigh made a crucial application of this formula that is often forgotten. While Equation (3.61) would apply to any small pattern of lines and spaces, Lord Rayleigh essentially looked at the extreme case of the smallest pitch that could be imaged – the resolution limit. The smallest pitch that can be printed would put the first diffracted order at the largest angle that could pass through the lens, defined by the numerical aperture, NA . For this one pattern, the general expression [Equation (3.61)] becomes the more familiar and specific Rayleigh DOF criterion:

$$DOF = k_2 \frac{\lambda}{NA^2} \quad (3.62)$$

The Rayleigh depth of focus criterion, as usually expressed in Equation (3.62), is rife with caveats. It was derived for a pattern of equal lines and spaces imaged under coherent

illumination, assumed that the feature was at the resolution limit, and made the paraxial approximation, which is not very accurate at numerical apertures in common lithographic use. In general, it does a very poor job of predicting the DOF observed in lithographic tools.

3.5 Imaging with Scanners Versus Steppers

In general, increasing the numerical aperture of a lens while keeping the field size constant requires significant improvements in lens design and manufacturing in order to keep aberration levels low (or better yet, to reduce them). A lens with a smaller field size will always be easier to design and manufacture. However, the field size must be at least the size of the largest chip to be manufactured, and larger field sizes provide for higher imaging productivity since throughput is improved if multiple chips can be imaged at once. This tension between smaller field sizes to improve aberration performance and larger field sizes to improve manufacturing productivity has led to a transition from steppers to *step-and-scan* tools.

A step-and-repeat camera (stepper) images a single field (for example, 20×20 mm in size on the wafer) while the mask and wafer are stationary, then steps the wafer to a new location and repeats the exposure. Eventually the whole wafer is covered with exposed fields. A scanner, on the other hand, will scan the mask and wafer simultaneously past an illuminated exposure field (in the shape of a slit), allowing for an exposure area limited only by the length of the slit and the travel of the mask and wafer. Step-and-scan tools are a hybrid between these two approaches. The mask and wafer are scanned past a small slit (for example, 26×8 mm) until the full mask is exposed (up to 26×33 mm at wafer dimensions). The wafer is then stepped to a new location and the scanning is repeated (Figure 3.22). The smaller image field (the slit) makes the imaging system simpler, though at the expense of more complicated mechanical mask and wafer motion. Thus, scanning can be considered a compromise between throughput, stage complexity and lens complexity for large-field imaging.

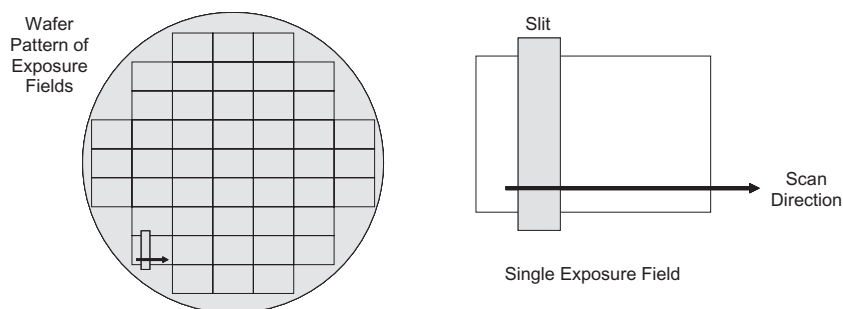


Figure 3.22 A wafer is made up of many exposure fields (with a maximum size that is typically 26×33 mm), each with one or more die. The field is exposed by scanning a slit that is about 26×8 mm across the exposure field

For scanners, exposure dose is controlled by the scan speed. Since step-and-scan tools generally use excimer laser light sources, which emit light in short pulses, the scan speed V needed to achieve a specific dose is given by

$$V = W_s \frac{f}{n} \quad (3.63)$$

where W_s is the slit width, f is the repetition rate of the laser and n is the number of pulses needed to achieve the proper dose. The pulse-to-pulse dose repeatability of a good excimer laser is a few percent. Since the overall dose repeatability is improved by a factor of $\sqrt{n}$ (assuming each pulse is independent), to achieve a dose control of a few tenths of a percent the number of pulses must be on the order of 100. For a slit width of 8 mm, a laser repetition rate of 2 kHz, and 50 pulses required, the wafer stage velocity would need to be 0.32 m/s (the reticle stage must move at exactly 4× this speed for a 4× reduction system). Improved laser repetition rates and reduced pulse-to-pulse variability allow for faster scan speeds and thus better throughput.

Besides all of the practical differences between the operation of steppers and step-and-scan tools, there are differences in imaging performance as well. Lenses are fabricated to be circularly symmetric, while chips are necessarily rectangular in shape. Thus, different-sized rectangular fields are possible for a given diameter lens. If the field size has a diameter of 28 mm, the maximum square field (for a stepper, for example), is just under 20 mm square. The same lens, however, can accommodate a 26×8 mm slit in a step-and-scan configuration, allowing for larger die sizes and/or more productive printing of multiple die in the larger field. In addition, the lens can be rotated during testing and assembly to find the 26×8 mm slice of the circular field that provides the best imaging quality.

As was discussed in sections 3.1 and 3.3, the nonideal imaging behavior of a lens (aberrations, flare) varies as a function of field position. The result is an unwanted variation in the dimensions and placements of the printed patterns as a function of field position. Scanning can significantly change the nature of this spatial variation across the field. When scanning the mask and wafer across the stationary slit, each point on the wafer is exposed by an average of aerial images – one for each field position along the scan direction of the slit. In other words, any variations in aberrations or flare across the field in the scan direction will be averaged out. Thus, the variations in feature size and overlay are significantly reduced in the scan direction. Of course, in the slit direction there is no such averaging and so the full spatial variation of lithographic results will remain in this direction (see Chapter 8 for a more detailed discussion of spatial CD variations).

The beneficial effects of aberration averaging mean that step-and-scan tools generally exhibit improved CD control compared to steppers. However, scanners also introduce new sources of process variation that, if not properly controlled, could erase these CD uniformity benefits. Scanning the mask and wafer past the imaging slit requires extremely precise synchronization of the mask and wafer position during the scan. Scan synchronization errors act exactly like wafer vibrations during a stepper exposure. The result is a blurring of the projected image, sometimes described as ‘image fading’. While systematic synchronization errors are possible, random errors are more common and are characterized by a *moving standard deviation* (MSD) of the stage position (see Figure 3.23). The

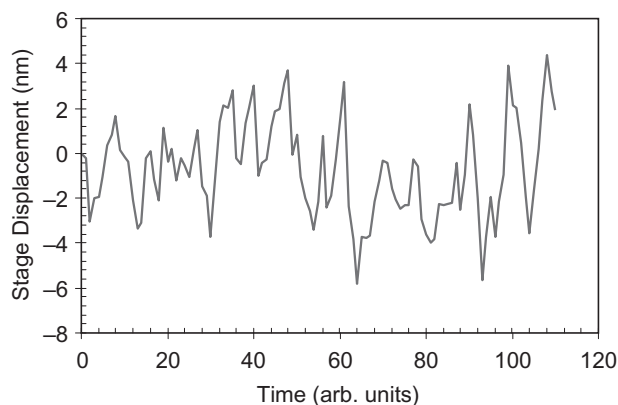


Figure 3.23 Example stage synchronization error (only one dimension is shown), with an MSD of 2.1 nm

resulting ‘averaged’ image is the convolution of the static image with a Gaussian of this standard deviation. Additionally, focus variations during the scan produce an averaging of the aerial image through focus, which will degrade image quality at best focus.

3.6 Vector Nature of Light

As James Clerk Maxwell has shown us, light is an electromagnetic wave. Electric and magnetic fields oscillate at some characteristic frequency while traveling at the speed of light. These fields have a magnitude, phase and direction. It is the direction (also called the polarization) of the electric field that defines its vector nature, and a key property of an electromagnetic wave is that its electric and magnetic field vectors are at right angles to each other, and at right angles to the direction of propagation. Figure 3.24 shows a standard textbook picture of a traveling ray of electromagnetic energy.

The direction of the electric field plays a critical role in the phenomenon of interference. Two plane waves, approaching a wafer at different angles, will interfere to form fringe patterns of light and dark making the simplest of lithographic patterns – an array of lines and spaces. This phenomenon of interference is the key to pattern formation. Without it, light hitting the wafer from different directions would simply add up to give a uniform intensity. We can see mathematically the effect of interference by examining how two electric fields combine to form a resultant electric field and intensity. Ignoring the refractive index of the medium, the intensity I is the square of the magnitude of the electric field E . If two electric fields are combined, what is the intensity of the combination? If the two electric fields do not interfere, the total intensity is the sum of the individual intensities.

$$I = |E_1|^2 + |E_2|^2 \quad (3.64)$$

If, however, the two electric fields interfere completely, the total intensity will be

$$I = |E_1 + E_2|^2 \quad (3.65)$$

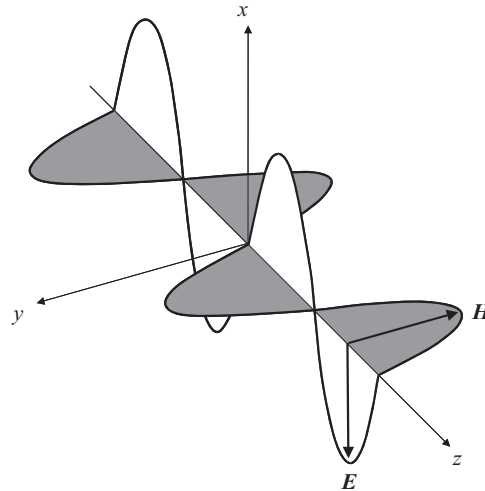


Figure 3.24 A monochromatic plane wave traveling in the z -direction. The electric field vector is shown as $\mathbf{E}$ and the magnetic field vector as $\mathbf{H}$

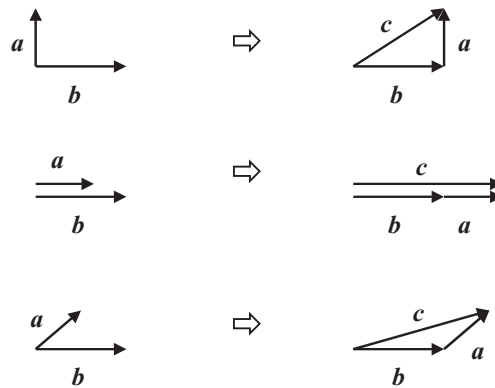


Figure 3.25 Examples of the sum of two vectors $\mathbf{a}$ and $\mathbf{b}$ to give a result vector $\mathbf{c}$, using the geometric ‘head-to-tail’ method

Two electric fields interfere with each other only to the extent that their electric fields oscillate in the same direction. If the electric fields are at right angles to each other, there will be no interference. Thus, to determine the amount of interference between two electric fields, one must first determine the amount of directional overlap between them. Vector mathematics gives us some simple tools to calculate directional overlap and thus the amount of interference. Figure 3.25 shows the standard head-to-tail method of geometrically adding two vectors. In the top case, the two vectors are at right angles to each other, so the head-to-tail construction forms a simple right triangle. Thus, the length of the resultant vector is given by the familiar Pythagorean theorem: $c^2 = a^2 + b^2$. If the

vector lengths represent the magnitudes of the electric field, the square of the length is the intensity and the resultant intensity is identical to that given by Equation (3.64). In other words, when the two electric fields are at 90° with respect to each other, they do not interfere and the total intensity is just the sum of the individual intensities.

At the other extreme (the middle example in Figure 3.25), two vectors in the same direction add directly and the resultant length is $c = a + b$. Thus, the intensity of the sum is given by Equation (3.65) and when the two electric fields are in the same direction, they interfere completely. Finally, the bottom example in Figure 3.25 shows an intermediate case. Working through the trigonometry of the vector sum shows that the portion of the two vectors that overlap will interfere, and the portion of the vectors that are at right angles will add in quadrature (i.e. they won't interfere, just add as intensities).

When thinking of light as a scalar rather than a vector quantity, we ignore the subtleties discussed above. Essentially, a scalar description of light always assumes that the electric fields are 100% overlapped and all electric fields add together as in Equation (3.65). Since interference is what gives us the patterns we want, a scalar view of light is too optimistic. The vector description of light says that, like the bottom case in Figure 3.25, there is usually some fraction of the electric fields of our two vectors that don't overlap and thus don't contribute to interference. The noninterfering light is still there, but it adds as a uniform intensity that degrades the quality of the image.

Consider the interference of two plane waves approaching the wafer at moderate angles, as in Figure 3.26. How will the two electric fields interfere? The electric field can point in any direction perpendicular to the direction of propagation. This arbitrary direction can in turn be expressed as the sum of any two orthogonal (basis) directions. The most convenient basis directions are called transverse electric or TE (the electric field pointing out of the page of the drawing) and transverse magnetic or TM (the electric field pointing in the page of the drawing). As Figure 3.26 shows, the TE case means that the electric fields of the two plane waves are always 100% overlapped regardless of the angle between the plane waves. For the TM case, however, the extent of overlap between the two vectors grows smaller as the angle between the plane waves grows larger. TM is the 'bad' polarization in lithography, especially at the very high numerical apertures that allow large angles between light rays.

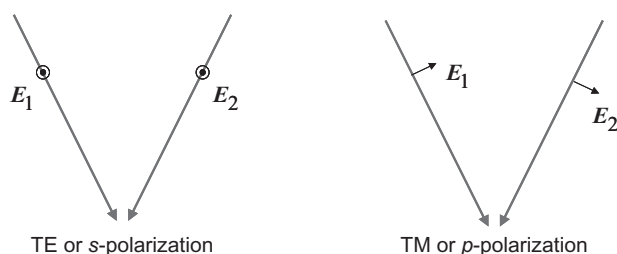


Figure 3.26 Two plane waves with different polarizations will interfere very differently. For transverse electric (TE) polarization (electric field vectors pointing out of the page), the electric fields of the two vectors overlap completely regardless of the angle between the interfering beams

We can calculate how much the two electric fields for the TM case will interfere with each other. Suppose that the two rays in Figure 3.26 are traveling at an angle θ with respect to the vertical direction (i.e. the direction normal to the wafer). The electric fields E_1 and E_2 will have an angle between them of 2θ . The amount of the electric field vector E_2 that points in the same direction as E_1 is just the geometric projection, $E_2 \cos(2\theta)$. Thus, the intensity will be given by the coherent (electric field) sum of the parts that overlap plus the incoherent (intensity) sum of the parts that don't overlap.

$$I = |E_1 + E_2 \cos(2\theta)|^2 + |E_2 \sin(2\theta)|^2 \quad (3.66)$$

Note that for $\theta = 0$, this equation reverts to the perfectly coherent (interfering) sum of Equation (3.65). For $\theta = 45^\circ$, the two electric fields are perpendicular to each other and Equation (3.66) becomes the perfectly incoherent (noninterfering) sum of Equation (3.64). For typical photoresists (with refractive indices between 1.6 and 1.7), this 45° angle inside the resist will be obtained for numerical apertures in the 1.15–1.2 range.

3.6.1 Describing Polarization

As we saw above, the behavior of a wave of light depends not only on the amplitude and phase of the wave (its scalar properties), but on the direction that the electric field is pointing (its vector property). In particular, the amount of interference between two electromagnetic waves depends on the relative directions of the two electric fields. If the electric fields point in the same direction, they will interfere completely. If they are orthogonal to one another, there will be no interference. The direction of the electric field of an electromagnetic wave, and how that direction varies in time or space, is called its *polarization*. Recall that Maxwell's equations require that the electric and magnetic fields are always perpendicular to each other and both always point perpendicular to the direction of propagation. As a result, it is sufficient to consider only the direction of the electric field.

Since an electromagnetic wave travels through time and space, there are three important ways of looking at (or thinking about) the direction of the electric field. First, imagine 'riding the wave': a given point on the wave moves through time and space at the speed of light. On this point on the wave, the electric field stays pointing in a constant direction, regardless of the type of polarization that the wave has. While an important conceptual idea, this way of thinking about the wave doesn't give us any information about its polarization. The second approach is to look at an instant of time, and see how the electric field direction is changing through space (this works best for simple harmonic waves, such as a plane wave). Finally, the third viewpoint is a fixed location in space, monitoring the change in the direction of the electric field through time. These last two perspectives allow for equivalent descriptions of polarization: the variation of the electric field direction in space at an instant in time, or the variation of the electric field direction in time at a fixed point in space.

With the above perspectives in mind, how can light be polarized? The simplest type of polarization is called *linear* polarization, shown in Figure 3.27. At an instant in time, the electric field E is always pointing in one direction (in this case, the y -direction, though in both positive and negative directions) for all points in space (Figure 3.27a). At one point in space, the electric field changes magnitude sinusoidally through time, but

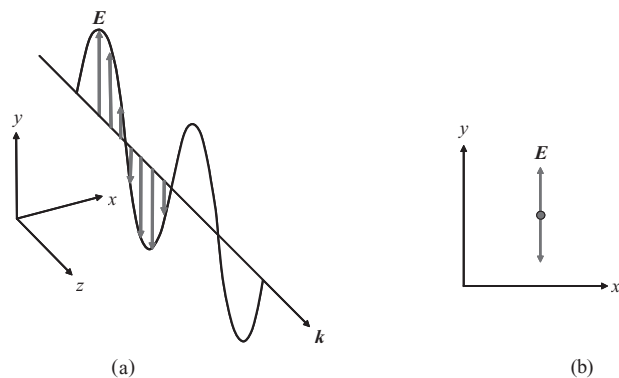


Figure 3.27 Linear polarization of a plane wave showing (a) the electric field direction through space at an instant in time, and (b) the electric field direction through time at a point in space. The $\mathbf{k}$ vector points in the direction of propagation of the wave

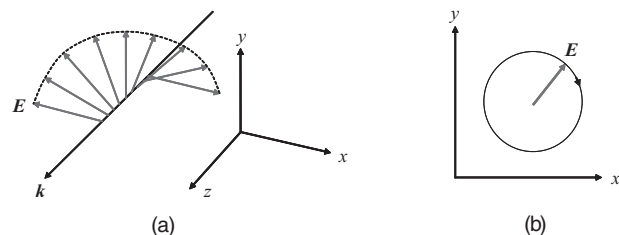


Figure 3.28 Right circular polarization of a plane wave showing (a) the electric field direction through space at an instant in time, and (b) the electric field direction through time at a point in space

always points in the same direction (Figure 3.27b). A second type of polarization is called *circular* polarization, as shown in Figure 3.28. Here, the electric field direction takes on a spiral form through space at a given instant in time (Figure 3.28a). At a given point in space, the electric field direction rotates through a circle, keeping a constant magnitude (Figure 3.28b). Both right-circular and left-circular polarization are possible, depending on the direction of rotation of the polarization. Note that the name given to the polarization comes from the shape traced out by the electric field through time at a given point in space. It is this viewpoint that is usually the easiest one to picture and understand.

Figure 3.29 compares some common types of polarization. Besides linear and circular polarization, *elliptical* polarization is somewhere in between the two. *Random* polarization is by far the most common type of polarization available for light sources, and until recently the only polarization available to lithographers. (Note that the common name for this type of polarization, unpolarized light, is actually a misnomer since light always has a polarization, even if it is random.) Here, the electric field direction changes randomly

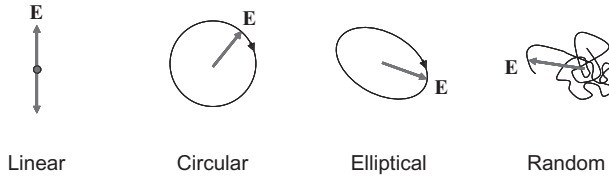


Figure 3.29 Examples of several types of polarizations (plotting the electric field direction through time at a point in space)

(and quickly) through time. When two waves of random polarization interfere with each other, some statistical average of amounts of electric field overlap must be used. Fortunately, a rigorous analysis provides a simple result: the intensity resulting from the interference of two randomly polarized waves can be calculated by breaking each wave into two linearly polarized orthogonal waves (for example, TE and TM polarizations), interfering each polarization separately, and summing the two resulting intensities.

In reality, none of the polarizations described above are achieved perfectly in practice. Practical light sources almost always produce some combination of polarized light and unpolarized light. To quantify the degree to which a light source supplies properly polarized illumination, a metric called *degree of polarization* (DOP) is used:

$$DOP = \frac{I_{\text{good pol}} - I_{\text{bad pol}}}{I_{\text{good pol}} + I_{\text{bad pol}}} \quad (3.67)$$

where $I_{\text{good pol}}$ is the desired polarization and $I_{\text{bad pol}}$ is the undesired polarization. If one is trying to make linear TE polarized light, for example, the good polarization is TE and the bad polarization is TM. Perfectly polarized light would have a DOP of 1.0. Randomly polarized light, being made up of equal parts TE and TM on average, will have a DOP of zero. Polarized light sources for use in lithography are expected to have DOPs ≥ 0.9 .

Mathematically, the polarization of a plane wave can be characterized by the phase difference between orthogonal components of the electric field. For example, consider a plane wave traveling in the z -direction with electric field components in the x - and y -directions. If the difference in phase between the x - and y -components of the electric field ($\Delta\phi$) is a multiple of π , the plane wave will be linearly polarized. If $\Delta\phi$ is an odd multiple of $\pi/2$, the wave will be circularly polarized. For other fixed phase differences, the wave will be elliptically polarized.

3.6.2 Polarization Example: TE Versus TM Image of Lines and Spaces

For TE illumination of line/space patterns, the electric fields of each of the diffracted orders overlap completely so that the vector sum of the electric fields equals the scalar sum. An example of a coherent illumination aerial image where only the zero and first diffracted orders are used was given in Chapter 2. If the lines are oriented in the y -direction, the only electric field component for this polarization will be in the y -direction as well:

$$E_y(x) = a_0 + 2a_1 \cos(2\pi x/p) \quad (3.68)$$

giving an intensity image of

$$I(x) = |E_y(x)|^2 = a_0^2 + 2a_1^2 + 4a_0a_1 \cos(2\pi x/p) + 2a_1^2 \cos(4\pi x/p) \quad (3.69)$$

A similar expression for TM illumination can also be derived. Here, the electric field will have both x and z components. If the diffraction angle of the first diffracted orders is given by $\sin\theta = \lambda/p$,

$$\begin{aligned} E_x &= a_0 + 2a_1 \cos\theta \cos(2\pi x/p) \\ E_z &= 2a_1 \sin\theta \sin(2\pi x/p) \end{aligned} \quad (3.70)$$

giving

$$\begin{aligned} I(x) &= |E_x(x)|^2 + |E_z(x)|^2 \\ &= a_0^2 + 2a_1^2 + 4a_0a_1 \cos\theta \cos(2\pi x/p) + 2a_1^2 \cos(2\theta) \cos(4\pi x/p) \end{aligned} \quad (3.71)$$

The effect of using the ‘bad’ polarization is to decrease the amount of interference by the cosine of the angle between the interfering orders. Thus, the interference between the zero and first orders is lessened by the factor $\cos\theta$ and the interference between the plus and minus first orders is decreased by $\cos 2\theta$. For randomly polarized light, the final aerial image would be the average of the two images given above for TE and TM polarization.

The difference between TE and TM polarization, as shown in the above example, can be quite large for small-pitch patterns. However, when considering only the aerial image (that is, the image in air) this difference is greatly exaggerated. When the image propagates from the air to the resist, refraction will decrease the angle of the first orders inside the resist according to Snell’s law. The lower angles in the resist will have greater overlap of the TM electric fields and thus a better quality image. For example, if the first order diffraction angle in air is $\sin\theta = 0.9$, then $\cos\theta = 0.44$ and the interference with the zero order is less than 50% effective. However, in resist the first-order angle is reduced by the refractive index (typically about 1.7), so that $\sin\theta_{\text{resist}} = 0.53$ and $\cos\theta_{\text{resist}} = 0.85$. The image in resist will have far higher quality than the aerial image. And of course, it is the image in resist that actually exposes the resist, not the aerial image. This topic will be discussed in greater detail in Chapter 4.

3.6.3 Polarization Example: The Vector PSF

An interesting (and somewhat nonintuitive) consequence of the vector nature of light is that linearly polarized illumination will produce an asymmetric PSF.⁸ A cross section of the PSF in the direction of polarization is wider than in the perpendicular direction (30% greater for the case of $\text{NA} = 0.866$, see Figure 3.30).

3.6.4 Polarization Aberrations and the Jones Pupil

As discussed at the beginning of this chapter, aberrations are characterized by means of the pupil function, defined as the ratio of the electric field at the exit pupil to the electric field at the entrance pupil. When thinking of electric fields as decomposed into orthogonal polarization components (TE and TM being the most convenient decomposition), an interesting question arises: will the pupil function for TE light passing through the lens

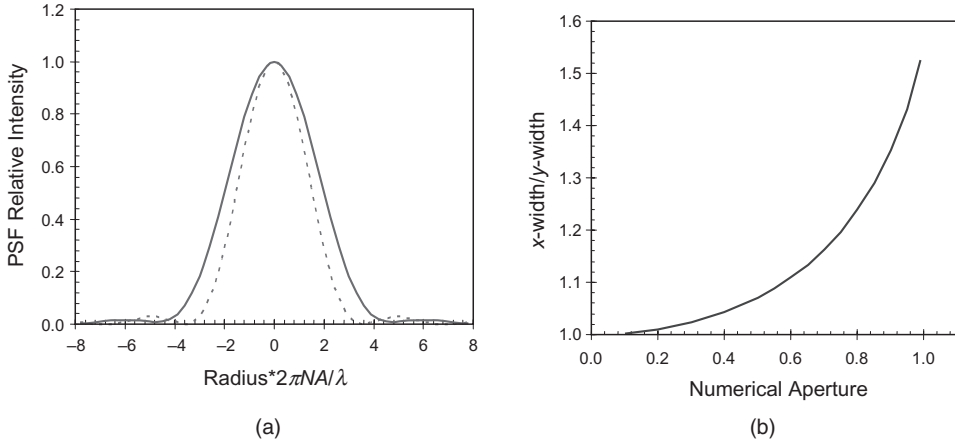


Figure 3.30 The point spread function (PSF) for linearly x-polarized illumination: (a) cross sections of the PSF for $NA = 0.866$ (solid line is the PSF along the x-axis, dashed line is the PSF along the y-axis); (b) ratio of the x-width to the y-width of the PSF as a function of numerical aperture

be the same as for TM light? Further, if TE light only enters the lens, will any of the light exiting the lens be TM? To include the variation of pupil transmittance with polarization, a more complete description of the pupil is required.

If the electric field entering the lens is broken down into TE and TM components, given by E_{iTE} and E_{iTM} , respectively, and the electric field at the exit pupil is given by E_{oTE} and E_{oTM} , these electric fields can be expressed as *Jones vectors*:⁹

$$\begin{bmatrix} E_{iTE} \\ E_{iTM} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} E_{oTE} \\ E_{oTM} \end{bmatrix} \quad (3.72)$$

Describing the transmittance of the lens by the *Jones pupil matrix* $\mathbf{J}$, the input and output Jones vectors are related by^{10,11}

$$\begin{bmatrix} E_{oTE} \\ E_{oTM} \end{bmatrix} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} E_{iTE} \\ E_{iTM} \end{bmatrix} \quad (3.73)$$

Note that if $J_{12} = J_{21} = 0$ and $J_{11} = J_{22}$, then the Jones pupil description of the lens transmittance reverts to the scalar pupil description where $J_{11} = J_{22} = P$, the scalar pupil function. It is thus convenient to define the scalar pupil function as the average of the diagonal elements of the Jones pupil:

$$P = \frac{J_{11} + J_{22}}{2} \quad (3.74)$$

Using this definition of the scalar pupil, a reduced Jones pupil $\tilde{\mathbf{J}}$ can be defined by

$$\mathbf{J} = P\tilde{\mathbf{J}} = P \begin{bmatrix} \tilde{J}_{11} & \tilde{J}_{12} \\ \tilde{J}_{21} & \tilde{J}_{22} \end{bmatrix} \quad (3.75)$$

For a lens that exhibits no polarization dependence, the reduced Jones matrix is simply the identity matrix. For a reasonably well-behaved lens, then, $\tilde{J}_{11}$ and $\tilde{J}_{22}$ are about equal to 1, and $\tilde{J}_{12}$ and $\tilde{J}_{21}$ are about equal to 0.

There are two main physical causes for what are sometimes called polarization aberrations. When light hits an interface (such as the surface of a lens element) at an oblique angle, the transmittance of that interface is different for TE and TM polarizations (a topic that will be discussed in some detail in Chapter 4). Antireflection coatings reduce but do not eliminate this difference. Thus, any lens that is a collection of refractive optical elements can be thought of as a weakly polarizing optical component, since the overall transmittance of the lens will depend on the incoming polarization and will vary with spatial frequency (the outer edges of the lens showing the biggest differences). This difference in amplitude transmittance as a function of polarization is called *diattenuation*. (A similar but less commonly observed phenomenon in lenses is *dichroism*, the variation in the absorption of light by the lens as a function of polarization.)

The second major cause of polarization effects in a lens is *birefringence*, a difference in the refractive index of a material as a function of the direction of the electric field vector. Birefringence leads to the general observation of *retardance*, a difference in the phase of the transmitted light as a function of its polarization. Some materials, most notably certain crystals, exhibit intrinsic birefringence. Most optical materials become at least somewhat birefringent when stressed. Common optical materials used in lithographic lenses exhibit small amounts of both intrinsic and stress-induced birefringence at 193-nm wavelength, and relatively large amounts of birefringence at a wavelength of 157 nm. Lens antireflection coatings can also contribute to retardance. Birefringence is usually expressed as the OPD experienced by two orthogonal polarizations per unit length traveled through the material, with units of nm/cm.

A useful mathematic decomposition of the Jones pupil expresses this matrix as linear combination of Pauli spin matrices:

$$\tilde{J} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + a_1 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + a_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + a_3 \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (3.76)$$

The magnitudes of a_1 , a_2 and a_3 then indicate the deviation of the reduced Jones pupil from an identity matrix, and can be easily related to the elements of the reduced Jones pupil matrix:

$$a_1 = \frac{\tilde{J}_{11} - \tilde{J}_{22}}{2}, \quad a_2 = \frac{\tilde{J}_{12} + \tilde{J}_{21}}{2}, \quad a_3 = \frac{\tilde{J}_{12} - \tilde{J}_{21}}{-2i} \quad (3.77)$$

These (complex number) coefficients have important physical interpretations. The real parts of a_1 and a_2 describe the diattenuation of the lens along the axes and at 45° to the axes, respectively. The imaginary parts of a_1 and a_2 describe the retardance of the lens along the axes and at 45° to the axes, respectively. The coefficient a_3 describes the circularly directed diattenuation and retardance.

All of these various forms for describing polarization-dependent pupil transmission suffer from a common problem – the excess complication that comes from expanding a scalar pupil (with its two pupil maps) into a vector pupil (with eight pupil maps). The

reduced Jones pupil form of Equation (3.75) preserves the intuition and experience of the scalar pupil, but does not reduce the overall level of complexity. Recently, Geh formulated an alternate decomposition of the Jones pupil into a scalar pupil plus a partial polarizer (producing diattenuation), a rotator (that simply rotates the polarization state), and a retarder (giving birefringence).¹² The resulting decomposition is significantly more intuitive.

One should note that the Jones pupil cannot describe the phenomenon of *depolarization*, the decrease in the degree of polarization as light travels through a lens. Depolarization is generally due to scattering mechanisms – the same mechanisms that give rise to flare. However, in general the amount of scattering will be approximately the same for each polarization. Thus, the mechanism used to account for flare described in section 3.3 can be coupled with the Jones pupil description of the nonscattered light transmittance to give a very reasonable and accurate description of lithographic imaging. The flare (scattered light) reaching the wafer is then assumed to be randomly polarized, thus accounting for depolarization.

3.7 Immersion Lithography

Although the scientific principles underlying immersion lithography have been known for well over 100 years (immersion microscopes were in common use by 1880), only recently has this technology attracted widespread attention in the semiconductor industry. Despite this rather late start, the potential of immersion lithography for improved resolution and depth of focus has changed the industry's roadmap and is destined to extend the life of optical lithography to new, smaller limits.

From a practical perspective, immersion lithography at 193 nm uses ultrapure degassed water, with refractive index of 1.437, as the immersion liquid. A small puddle of water is kept between the lens and the wafer during wafer exposure (Figure 3.31). New water is constantly pumped under the lens to keep the optical properties of the fluid consistent, and to prevent the buildup of contaminants. A special wafer chuck keeps the puddle intact as the edge of the wafer is scanned under the puddle.

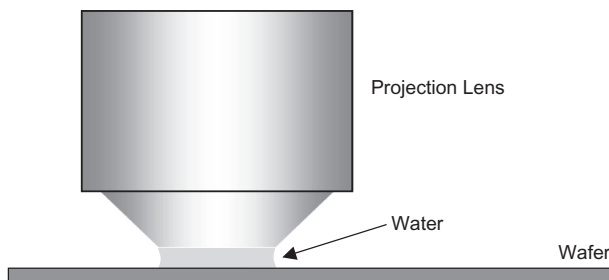


Figure 3.31 Immersion lithography uses a small puddle of water between the stationary lens and the moving wafer. Not shown is the water source and intake plumbing that keeps a constantly fresh supply of immersion fluid below the lens

3.7.1 The Optical Invariant and Hyper-NA Lithography

The story of immersion lithography begins with Snell's Law. Light traveling through material 1 with refractive index n_1 strikes a surface with angle θ_1 relative to the normal to that surface. The light transmitted into material 2 (with index n_2) will have an angle θ_2 relative to that same normal as given by Snell's law.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (3.78)$$

Now picture this simple law applied to a film stack made up of any number of thin parallel layers (Figure 3.32a). As light travels through each layer, Snell's law can be repeatedly applied:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3 = n_4 \sin \theta_4 = \dots = n_k \sin \theta_k \quad (3.79)$$

Thanks to Snell's law, the quantity $n \sin \theta$ is *invariant* as a ray of light travels through this stack of parallel films. Interestingly, the presence or absence of any film in the film stack in no way affects the angle of the light in other films of the stack. If films 2 and 3 were removed from the stack in Figure 3.32a, for example, the angle of the light in film 4 would be exactly the same.

We find another, related invariant when looking at how an imaging lens works. A well-made imaging lens (with low levels of aberrations) will have a *Lagrange invariant* (often just called the optical invariant) that relates the angles entering and exiting the lens to reduction ratio of that lens, as described in Chapter 2:

$$R = \frac{n_w \sin \theta_w}{n_m \sin \theta_m} \quad (3.80)$$

where n_m is the refractive index of the media on the mask side of the lens, θ_m is the angle of a ray of light entering the lens relative to the optical axis, n_w is the refractive index of the media on the wafer side of the lens and θ_w is the angle of a ray of light exiting the lens relative to the optical axis (Figure 3.32b). Note that, other than a scale factor given by the magnification of the imaging lens and a change in the sign of the angle to account for the focusing property of the lens, the Lagrange invariant makes a lens seem like a

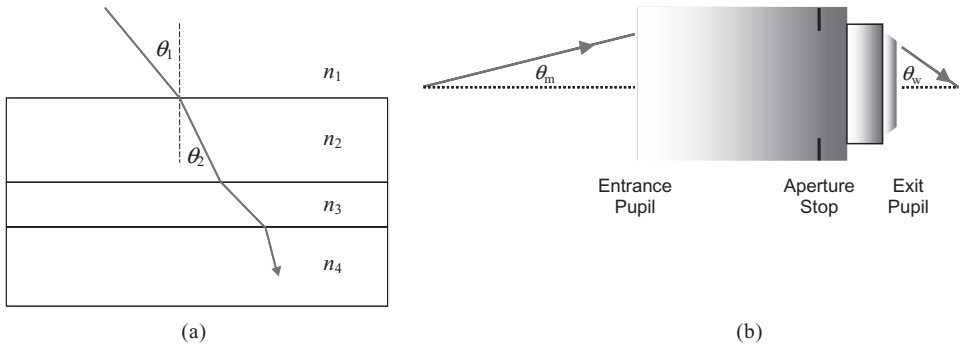


Figure 3.32 Two examples of an 'optical invariant', (a) Snell's law of refraction through a film stack, and (b) the Lagrange invariant of angles propagating through an imaging lens

thin film obeying Snell's law. (It is often convenient to imagine the imaging lens as $1\times$, scaling all the object dimensions by the magnification, thus allowing $R = 1$ and making the Lagrange invariant look just like Snell's law).

These two invariants can be combined when thinking about how a photolithographic imaging system works. Light diffracts from the mask at a particular angle. This diffracted order propagates through the lens and emerges at an angle given by the Lagrange invariant. This light then propagates through the media between the lens and the wafer and strikes the photoresist. Snell's law dictates the angle of that ray in the resist, or any other layers that might be coated on the wafer. Taking into account the magnification scale factor, the quantity $n\sin\theta$ for a diffracted order is constant from the time it leaves the mask to the time it combines inside the resist with other diffraction orders to form an image of the mask.

If we replace the air between the lens and the wafer with water, the optical invariant says that the angles of light inside the resist will be the same, presumably creating the exact same image. The impact of immersion lithography comes from two sources: the maximum possible angle of light that can reach the resist, and the phase of that light.

Consider again the chain of angles through multiple materials as given by Equation (3.79). Trigonometry will never allow the sine of an angle to be greater than 1. Thus, the maximum value of the invariant will be limited by the material in the stack with the smallest refractive index. If one of the layers is air (with a refractive index of 1.0), this will become the material with the smallest refractive index and the maximum possible value of the invariant will be 1.0. If we look then at the angles possible inside of the photoresist, the maximum angle possible would be $\sin\theta_{\text{max,resist}} = 1 / n_{\text{resist}}$. Now suppose that the air is replaced with a fluid of a higher refractive index, but still smaller than the index of the photoresist. In this case, the maximum possible angle of light inside the resist will be greater: $\sin\theta_{\text{max,resist}} = n_{\text{fluid}} / n_{\text{resist}}$. At a wavelength of 193 nm, resists have refractive indices of about 1.7 and water has a refractive index of about 1.44. The fluid does not make the angles of light larger, but it *enables* those angles to be larger. If one were to design a lens to emit larger angles, immersion lithography will allow those angles to propagate into the resist. The numerical aperture of the lens (defined as the maximum value of the invariant $n\sin\theta$ that can pass through the lens) can be made to be much larger using immersion lithography, with the resulting improvements in resolution one would expect.

Thus, by using an immersion fluid between the lens and the wafer, numerical apertures greater than 1 have been achieved. These so-called hyper-NA immersion lithography tools are certainly not the first $\text{NA} > 1$ imaging tools, since immersion microscopes have been common for over 100 years. The challenges in making immersion practical for high-volume semiconductor manufacturing, however, are quite great, especially considering the need for high-speed scanning of the wafer past the lens while immersed in the fluid. Practical numerical apertures for immersion systems are probably in the range of 93% of the refractive index of the fluid. For water, with $n = 1.44$ at 193 nm, this means a numerical aperture of up to 1.35. High-index fluids ($n = 1.65$, for example) could enable numerical apertures as high as 1.5. Beyond this level, the maximum angle is limited by the refractive index of the photoresist, which would have to be raised to allow for further increases in numerical aperture. From the lens perspective, the refractive index of the various lens materials for 193-nm lithography is on the order of 1.5, so that concave lens

elements would be required in order to couple light into a higher-index fluid. Alternatively, one could attempt to develop higher index glasses for use as lens elements. Both of these possibilities pose quite challenging problems.

The second way that an immersion fluid changes the results of imaging comes from manner in which the fluid affects the phase of the light as it reaches the wafer. Light, being a wave, undergoes a phase change as it travels. If light of (vacuum) wavelength λ travels some distance Δz through some material of refractive index n , it will undergo a phase change $\Delta\phi$ given by

$$\Delta\phi = 2\pi n\Delta z/\lambda \quad (3.81)$$

A phase change of 360° will result whenever the optical path length (the refractive index times the distance traveled) reaches one wavelength. This is important in imaging when light from many different angles combine to form one image. All of these rays of light will be in phase only at one point – the plane of best focus. When out of focus, rays traveling at larger angles will undergo a larger phase change than rays traveling at smaller angles. As a result, the phase difference between these rays will result in a blurred image.

For a given diffraction order (and thus a given angle of the light inside the resist), the angle of the light inside an immersion fluid will be less than if air were used. These smaller angles will result in smaller OPDs between the various diffracted orders when out of focus, and thus a smaller degradation of the image for a given amount of defocus. In other words, for a given feature being printed and a given numerical aperture, immersion lithography will provide a greater depth of focus (DOF). A more thorough description of the impact of immersion on DOF will be given in the following section.

3.7.2 Immersion Lithography and the Depth of Focus

From the derivation of the Rayleigh DOF criterion given above in section 3.4.7, we can state the restrictions on this conventional expression of the Rayleigh DOF: relatively low numerical apertures imaging a binary mask pattern of lines and spaces at the resolution limit. Equation (3.60) can be thought of as the high-NA version of the Rayleigh DOF criterion, which still assumes we are imaging a small binary pattern of lines and spaces, but is appropriate at any numerical aperture. It also accounts for immersion lithography through the refractive index n . The angle θ can be related to the pitch by letting this angle be that of the first diffraction order.

$$n \sin \theta = \frac{\lambda}{p} \quad (3.82)$$

Combining Equations (3.60) and (3.82), one can see how immersion will improve the DOF of a given feature:

$$\frac{DOF(immersion)}{DOF(dry)} = \frac{1 - \sqrt{1 - (\lambda/p)^2}}{n - \sqrt{n^2 - (\lambda/p)^2}} \quad (3.83)$$

As Figure 3.33 shows, the improvement in DOF is at least the refractive index of the fluid, and grows larger from there for the smallest pitches.

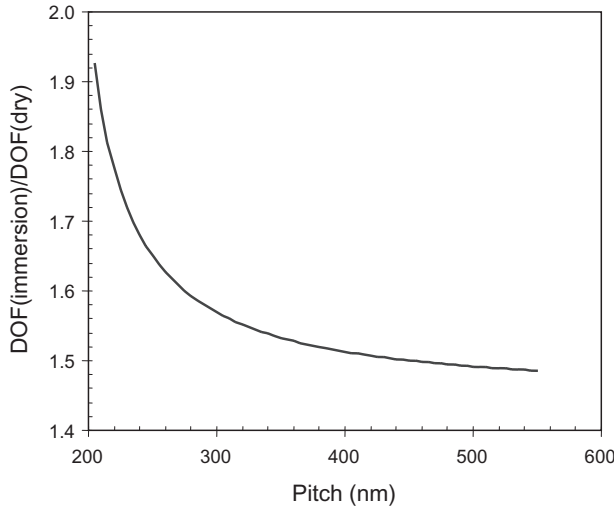


Figure 3.33 For a given pattern of small lines and spaces, using immersion improves the depth of focus (DOF) by at least the refractive index of the fluid (in this example, $\lambda = 193 \text{ nm}$, $n_{\text{fluid}} = 1.46$)

3.8 Image Quality

While much of this chapter has focused on the nature of imaging and those factors that affect the resulting aerial image, along the way we have had occasion to mention the ‘quality’ of the aerial image without actually defining what is meant by this term. While it is intuitively easy to see how defocus and aberrations degrade the image, we will need to define a numerical metric of image quality in order to quantify these effects. While much more will be said on this topic in Chapter 9, here we will simply define a metric related to the ability of the image to define and control the position of the edge to be printed in photoresist.

3.8.1 Image CD

The aerial image, when propagated into a photoresist, interacts with the resist to create a final feature of a certain width. It is often convenient to estimate the width of the final resist feature by the width of the aerial image. Consider an ideal, infinite contrast positive resist. For such a resist, any exposure above some threshold dose E_{th} will cause all of the resist to be removed. Thus, for an aerial image $I(x)$ and an exposure dose E , the critical dimension (CD) will equal $2x$ when

$$E_{\text{th}} = EI(x) \quad (3.84)$$

For a given dose, an image intensity threshold value can be defined as

$$I_{\text{th}} = \frac{E_{\text{th}}}{E} \quad (3.85)$$

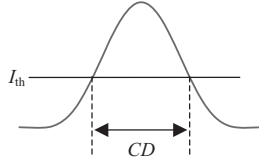


Figure 3.34 Defining image CD: the width of the image at a given threshold value I_{th}

The image CD is simply the width of the image evaluated at this intensity threshold (Figure 3.34).

This simple threshold model for estimating the resist CD can be significantly improved by including a ‘resist bias’. As Figure 3.20 shows and section 3.4.4 discusses, aerial images exhibit an isofocal point, an x -position where the intensity varies the least with defocus. Likewise, resist CD exhibits an isofocal value (as will be discussed in Chapter 8) where the CD is least sensitive to changes in focus. In general, the image isofocal point will not occur at the same position as the resist isofocal point. The difference between these points is called the *resist bias*.

Our simple threshold model can be modified to account for this empirical observation by adding a bias to Equation (3.84):

$$I_{th} = I(x + b) \quad (3.86)$$

where b is called the edge bias and the CD is still $2x$ for a symmetrical image. It should be evident that the threshold intensity and the resist bias are not independent parameters, but are related by the slope of the aerial image.

$$\frac{dI_{th}}{db} = \left. \frac{dI}{dx} \right|_{x+b} \quad (3.87)$$

Integrating this relationship,

$$I_{th} = I_{th0} + \frac{dI}{dx} b \quad (3.88)$$

where I_{th0} is the threshold one would obtain if the bias were zero.

There are still only two independent parameters in Equation (3.88). If I_{th0} and b are used as the parameters, then the threshold intensity used to extract the edge position from the image will vary as a function of the image slope. This is a simple form of a *variable threshold resist model*. Adding a dose dependence, as in Equation (3.85), we obtain

$$I(x = CD/2) = I_{th} = \frac{E_{th0}}{E} + \frac{dI}{dx} b \quad (3.89)$$

where the slope is evaluated at $I = I_{th}$. Since this requires knowing I_{th} before evaluating the slope, either an iterative procedure must be used or one must assume that the slope is constant over the range of values of b commonly encountered. A second form of the

model avoids this complication by using I_{th0} (or E_{th0}) and I_{th} as the parameters. In this case, the bias will vary as a function of the image slope and the result will be a *variable bias resist model*. For the variable bias resist model,

$$I_{th} = I(x + b), \quad b = \frac{I_{th} - I_{th0}}{dI/dx} \quad (3.90)$$

where the slope is now evaluated at a constant $I = I_{th}$.

3.8.2 Image Placement Error (Distortion)

The threshold model used above to define the width of an aerial image can also be used to define its center position, called the image placement. Since this center position is the average of the two edge positions, a constant bias does not impact image placement. However, a variable threshold or variable bias resist model, as discussed above, will influence the center position whenever the image slope is different on the right and left sides of the image.

Any deviation in the image placement from the ideal (desired) position is called *image placement error* (as described above in section 3.1.3). When image placement error does not depend on the size or surroundings of the feature in question, this error is called *distortion*. The most common cause of distortion is the aberration of tilt. Variation of image placement error with feature size or type can be caused by any odd Zernike term other than tilt.

3.8.3 Normalized Image Log-Slope (NILS)

The slope of the image intensity I as of function of position x (dI/dx) measures the steepness of the image in the transition from bright to dark. However, for the slope to be useful as a metric of image quality, the intensity must be normalized by dividing the slope by the intensity at the point where the slope is being measured. The resulting metric is called the image log-slope (ILS):

$$Image \text{ Log - Slope} = \frac{1}{I} \frac{dI}{dx} = \frac{d \ln(I)}{dx} \quad (3.91)$$

where this log-slope is measured at the nominal (desired) line edge (Figure 3.35). Since variations in the photoresist edge positions (linewidths) are typically expressed as a percentage of the nominal linewidth, the position coordinate x can also be normalized by multiplying the log-slope by the nominal linewidth w , to give the normalized image log-slope (NILS).

$$NILS = w \frac{d \ln(I)}{dx} \quad (3.92)$$

The normalized image log-slope can now be used to calculate the image quality of various aerial images that we have discussed in this chapter. Consider first a generic 1D image of a line-space pattern of pitch p . This image can be represented as a Fourier series

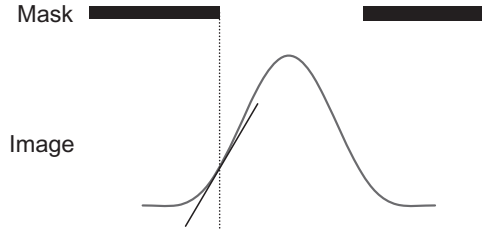


Figure 3.35 Image Log-Slope (or the Normalized Image Log-Slope, NILS) is the best single metric of image quality for lithographic applications

of some number of terms N (typically equal to twice the highest diffraction order number that passes through the lens).

$$I(x) = \sum_{j=0}^N \beta_j \cos(2\pi jx/p) \quad (3.93)$$

The NILS calculated at the edge position $x = w/2$ will be

$$NILS = - \left(\frac{2\pi w}{p} \right) \frac{\sum_{j=0}^N j \beta_j \sin(\pi jw/p)}{\sum_{j=0}^N \beta_j \cos(\pi jw/p)} \quad (3.94)$$

For the special case of equal lines and spaces, so that $w = p/2$,

$$NILS = -\pi \frac{\sum_{j=0}^N j \beta_j \sin(\pi j/2)}{\sum_{j=0}^N \beta_j \cos(\pi j/2)} = -\pi \frac{\beta_1 - \beta_3 + \beta_5 - \dots}{\beta_0 - \beta_2 + \beta_4 - \dots} \quad (3.95)$$

For small-pitch patterns, only a few diffraction orders make it through the lens. Near the resolution limit, where the first diffraction orders are the highest ones transmitted through the lens, $N = 2$ and

$$NILS = -\pi \frac{\beta_1}{\beta_0 - \beta_2} \quad (3.96)$$

One can see from this simple derivation that a high-magnitude NILS results from larger amounts of high-order components (larger β_1 and β_2) and less zero-order component (smaller β_0).

Note that NILS can be positive or negative, depending on whether the pattern is a line or a space, and whether the right or the left edge of the pattern is being considered. As a convention, the sign of the NILS expression is usually adjusted to produce a positive value of NILS. Table 3.2 shows the calculated value of NILS for each of the equal line/space aerial images discussed in this chapter. Of the images shown, the coherent, in-focus

Table 3.2 Calculation of NLS for various equal line/space images presented in this chapter

Image Type	Equation	β_0	β_1	β_2	NLS
Coherent, in focus, TE	(3.69)	$\frac{1}{4} + \frac{2}{\pi^2}$	$\frac{2}{\pi}$	$\frac{2}{\pi^2}$	8
Coherent, in focus, TM	(3.71)	$\frac{1}{4} + \frac{2}{\pi^2}$	$\frac{2}{\pi} \cos \theta$	$\frac{2}{\pi^2} \cos(2\theta)$	$\frac{8 \cos \theta}{1 + \frac{8}{\pi^2} \sin^2 \theta}$
Coherent, out of focus, TE	(3.27)	$\frac{1}{4} + \frac{2}{\pi^2}$	$\frac{2}{\pi} \cos(\Delta\Phi)$	$\frac{2}{\pi^2}$	$8 \cos(\Delta\Phi)$
Partially coherent, out of focus, TE	(3.29)	$\frac{1}{4} + \frac{2}{\pi^2}$	$\frac{2}{\pi} \cos(\Delta\Phi) \frac{2J_1(a)}{a}$	$\frac{2}{\pi^2} \frac{J_1(2a)}{a}$	$\frac{8 \cos(\Delta\Phi) \frac{2J_1(a)}{a}}{1 + \frac{8}{\pi^2} \left(1 - \frac{J_1(2a)}{a}\right)}$
Incoherent, in focus, TE	See Chapter 2	$\frac{1}{2}$	$\frac{2}{\pi} MTF(1/p)$	0	$4MTF(1/p)$

image from TE illumination produces the highest NLS (equal to 8). Defocus or the use of TM illumination causes a decrease in the NLS, and thus a degradation of image quality.

In Chapter 2, the Kintner method was used to calculate the partially coherent in-focus image of small lines and spaces as the weighted sum of one-beam, two-beam and three-beam images. This Kintner formulation can also be put into the Fourier series form of Equation (3.93). For TE illumination where only the zero and $\pm$ first diffraction orders are used,

$$\begin{aligned}
 \beta_0 &= a_0^2 + a_1^2(s_2 + 2s_3) \\
 \beta_1 &= 2a_0a_1(s_2 + 2s_3) \\
 \beta_2 &= 2a_1^2(s_3)
 \end{aligned} \tag{3.97}$$

where s_2 and s_3 are the fractions of two-beam and three-beam imaging, respectively, as given in Tables 2.3 or 2.4.

3.8.4 Focus Dependence of Image Quality

All of the image quality metrics described above – image CD, image placement and image log-slope – will in general vary with focus. The variation of image CD with focus is sometimes called isofocal bias, a topic that will be discussed at length in Chapter 8. The variation of image log-slope with focus is a common focus characterization technique, and is discussed extensively in Chapter 9. The variation of image placement with focus is called telecentricity error, and is commonly due to asymmetric source shapes (discussed briefly in Chapter 8).

Problems

- 3.1. Derive Equation (3.5).
- 3.2. Equation (3.7), which was derived for coherent illumination, predicts that coma will result in a positive placement error for lines and spaces when the pitch is less than $\sqrt{1.5}\lambda/NA$, a negative placement error for pitches greater than this amount, and no placement error when the pitch equals $\sqrt{1.5}\lambda/NA$. Show why this is true using pictures of the wavefront error and the diffraction pattern.
- 3.3. Using the Zernike polynomial term F_4 , explain the effect of 3rd-order astigmatism on x -oriented lines and spaces versus y -oriented lines and spaces.
- 3.4. Calculate the RMS wavefront error, W_{RMS} , and the resulting Strehl ratio for the case when
 - (a) Only 3rd-order spherical aberration is present
 - (b) Only 3rd-order x -coma is present

Note that

$$W_{\text{RMS}}^2 = \frac{1}{\pi} \int_0^1 \int_0^{2\pi} W^2(R, \phi) R dR d\phi$$

- 3.5. Consider coherent imaging of equal lines and spaces where only the zero and $\pm$ first diffraction orders enter the lens. Derive an expression, similar to Equation (3.24), that shows how spherical aberration results in a pitch-dependent paraxial focus error.
- 3.6. Calculate the impact of long-range flare on the image log-slope.
- 3.7. Consider the case of dense equal lines and spaces (only the 0 and ± 1 st orders are used) imaged with coherent illumination. Show that the peak intensity of the image in the middle of the space falls off approximately quadratically with defocus for small amounts of defocus.
- 3.8. Using the geometry of Figure 3.18b, derive Equation (3.20).
- 3.9. Compare the depth of focus predictions of the high-NA version of the Rayleigh equation [Equation (3.60)] to the paraxial version of Equation (3.61) by plotting predicted DOF versus pitch (use $k_2 = 0.6$, $\lambda = 248$ nm, pitch in the range from 250 to 500 nm, and assume imaging in air).
- 3.10. Consider two unit-amplitude plane waves traveling in air at angles $\pm\theta$ with respect to the z -axis (as in Figure 3.26). Show that the resulting interference patterns, depending on the polarizations of the two beams, will be

$$\begin{aligned} \text{TE-polarization: } I(x) &= 2 + 2 \cos(4\pi x \sin(\theta) / \lambda) \\ \text{TM-polarization: } I(x) &= 2 + 2 \cos(2\theta) \cos(4\pi x \sin(\theta) / \lambda) \end{aligned}$$

What is the pitch of the resulting line/space image?

- 3.11. Derive Equation (3.83). What is the limit of this equation for $p \gg \lambda$?
- 3.12. Consider a coherent, three-beam image of a line/space pattern under TE illumination ($NA = 0.93$, $\lambda = 193$ nm, linewidth = 100 nm, pitch = 250 nm). Plot the resulting aerial image at best focus and at defocus values of 50 and 100 nm for:
 - (a) imaging in air
 - (b) immersion imaging ($n_{\text{fluid}} = 1.44$)

- 3.13. Consider an in-focus, TE image where only the zero order and one of the first orders pass through the lens. For this image, what are the values of β_0 , β_1 and β_2 in Equation (3.93)? What is the value of the NILS for the case of equal lines and spaces?
- 3.14. For the line/space image of Equation (3.93):
- derive an equation for the image contrast, defined as

$$\text{Image Contrast} = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$$

For simplicity, assume the maximum intensity occurs in the middle of the space, and the minimum intensity occurs in the middle of the line.

- For the case where the second harmonic is the largest harmonic in the image (i.e. when $N = 2$), compare the resulting image contrast to the NILS of Equation (3.96).
- 3.15. Consider an in-focus, coherent, three-beam image of an equal line/space pattern under TE illumination ($NA = 0.93$, $\lambda = 193$ nm, linewidth = 130 nm, pitch = 260 nm). Use a threshold model ($E_{\text{th}} = 20$ mJ/cm²) to plot the resulting line CD versus dose. What dose produces the nominal feature size?

References

- Ulrich, W., Rostalski, H.-J. and Hudyma, R., 2004, Development of dioptric projection lenses for deep ultraviolet lithography at Carl Zeiss, *Journal of Microlithography, Microfabrication, and Microsystems*, **3**, 87–96.
- Kroyan, A., Bendik, J., Semprez, O., Farrar, N., Rowan, C. and Mack, C.A., 2000, Modeling the effects of excimer laser bandwidths on lithographic performance, *Proceedings of SPIE: Optical Microlithography XIII*, **4000**, 658–664.
- Lai, K., Lalovic, I., Fair, B., Kroyan, A., Progler, C., Farrar, N., Ames, D. and Ahmed, K., 2003, Understanding chromatic aberration impacts on lithographic imaging, *Journal of Microlithography, Microfabrication, and Microsystems (JM3)*, **2**, 105–111.
- Flagello, D.G. and Pomerene, A.T.S., 1987, Practical characterization of 0.5 μ m optical lithography, *Proceedings of SPIE: Optical Microlithography VI*, **772**, 6–20.
- Mack, C.A. and Kaufman, P.M., 1988, Mask bias in submicron optical lithography, *Journal of Vacuum Science & Technology B*, **B6**, 2213–2220.
- Lai, K., Wu, C. and Progler, C., 2001, Scattered light: the increasing problem for 193 nm exposure tools and beyond, *Proceedings of SPIE: Optical Microlithography XIV*, **772**, 1424–1435.
- Fukuda, H., Hasegawa, N., Tanaka, T. and Hayashida, T., 1987, A new method for enhancing focus latitude in optical lithography: FLEX, *IEEE Electron Device Letters*, **EDL-8**, 179–180.
- Richards, B. and Wolf, E., 1959, Electromagnetic diffraction in optical systems. II. Structure of the image field in an aplanatic system, *Proceedings of the Royal Society of London, Series A*, **253**, 358–379.
- Jones, R.C., 1941, New calculus for the treatment of optical systems, *Journal of the Optical Society of America*, **31**, 488–493.
- Totzeck, M., Gräupner, P., Heil, T., Göhnermeier, A., Dittmann, O., Krähmer, D., Kamenov, V., Ruoff, J. and Flagello, D., 2005, Polarization influence on imaging, *Journal of Microlithography, Microfabrication, and Microsystems*, **4**, p. 031108.
- McIntyre, G., Kye, J., Levinson, H. and Neureuther, A., 2006, Polarization aberrations in hyper-numerical aperture projection printing: a comparison of various representations, *Journal of Microlithography, Microfabrication, and Microsystems*, **5**, p. 033001.

- 12 Geh, B., Ruoff, J., Zimmermann, J., Gräupner P., Totzeck, M., Mengel, M., Hempelmann, U. and Schmitt-Weaver, E., 2007, The impact of projection lens polarization properties on lithographic process at hyper-NA, *Proceedings of SPIE: Optical Microlithography* **6520**, 65200F.